COMPLEX GEOMETRY

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Contents

1. Complex structures on vector spaces

Let's discuss complex structures at the level of linear algebra.

Definition 1.1. An *almost complex structure* on a vector space V is an endomorphism $I \in \operatorname{End}(V)$ such that $I^2 = -\mathrm{Id}_V$.

Lemma 1.2. *A vector space endowed with an almost complex structure admits a natural structure of a complex vector space.*

Proof. $(a+ib)v := av + bI(v)$. □

Conversely, given a complex vector space, multiplication by i produces an almost complex structure. Thus the two notions become the same.

Now we introduce the complexification of a real vector space. Let $V_{\mathbb{C}} := V \otimes_{\mathbb{R}} \mathbb{C}$. This means that elements of $V_{\mathbb{C}}$ are finite sums of elements of the form $v \otimes (a + iv)$ for $a, b \in \mathbb{R}$ and $i = \sqrt{-1}$.

Notice that although (V, I) is also a complex vector space, it is a different one from $V_{\mathbb{C}}$. The complexification $V_{\mathbb{C}}$ is much more interesting since it gives us the (p, q)-decomosition when we take the exterior algebra.

Further, $V_{\mathbb{C}} = V \oplus iV$. We define *conjugation* of vectors in $V_{\mathbb{C}}$ by $\bar{v} = \overline{v \otimes \lambda} := v \otimes \bar{\lambda}$

This means that $\overline{V}$ is just V but with the scalar product $\lambda v := \bar{\lambda} v$. for any $v \in V_{\mathbb{C}} := V \otimes_{\mathbb{R}} \mathbb{C}$. This makes the equation $\overline{\Lambda^{1,0}(V)} = \Lambda^{0,1}(V)$, which appears in the next proposition, obvious. (See [Huy05, p. 25].)

Lemma 1.3. *We have a decomposition*

$$V_{\mathbb{C}} = V^{1,0} \oplus V^{0,1}$$

given by the eigenspaces of I. $\overline{\Lambda^{1,0}} = \Lambda^{0,1}$ holds by definition of conjugation.

Now let's discuss the p, q decomposition. The upshot is the following:

$$T^2(V \oplus W) = T^2(V) \otimes T^0(W) \oplus T^1(V) \otimes T^1(W) \oplus T^0(V) \otimes T^2(W).$$

Explicitly, the decomposition of a general element on the LHS is

$$\begin{aligned}(v_1, w_1) \otimes (v_2, w_2) &= (v_1, 0) \otimes (v_2, w_2) + (0, w_1) \otimes (v_2, w_2) \\ &= (v_1, 0) \otimes (v_2, 0) + (v_1, 0) \otimes (0, w_2) \\ &+ (0, w_1) \otimes (v_2, 0) + (0, w_1) \otimes (0, w_2)\end{aligned}$$

which is valid because the tensor product of a vector space T^2X is just the pairs $x \otimes y$ subject to relations that $(x_1 + x_2) \otimes y = x_1 \otimes y + x_2 \otimes y$ and addition for y and scalar product. So we just identify elements

$$\begin{aligned}(v_1, 0) \otimes (v_2, 0) &\leftrightsquigarrow v_1 \otimes v_2 \in T^2(V) \otimes T^0(W) := T^{2,0}(V \oplus W) \\ (v_1, 0) \otimes (0, w_2) &\leftrightsquigarrow v_1 \otimes w_2 \in T^1(V) \otimes T^1(W) := T^{1,1}(V \oplus W) \\ (0, w_1) \otimes (v_2, 0) &\leftrightsquigarrow w_1 \otimes v_2 \in T^1(W) \otimes T^1(V) \cong T^1(V) \otimes T^1(W) := T^{1,1}(V \oplus W) \\ (0, w_1) \otimes (0, w_2) &\leftrightsquigarrow w_1 \otimes w_2 \in T^0(V) \otimes T^2(W) := T^{0,2}(V \oplus W).\end{aligned}$$

We have proved that any element in $T^2(V \oplus W)$ can be written as an element in the vector space $T^2(V) \otimes T^0(W) + T^1(V) \otimes T^1(W) + T^0(V) \otimes T^2(W)$. Since the intersection of the factors is disjoint, the sum can be taken direct.

Notice that we have two copies of $T^{1,1}(V \oplus W)$. By the binomial theorem and isomorphisms of the kind $A \otimes B \cong B \otimes A$, we readily see that

$$T^k(V \oplus W) \cong \bigoplus_{p+q=k} \binom{k}{p} T^p(V) \otimes T^q(W).$$

Of course the binomial coefficient matters little when the underlying field is $\mathbb{C}$.

The next step is to take the quotient on the LHS by the relation $v \otimes v = 0$, i.e., the ideal generated by $v \otimes v$. When we compute $(v_1, w_1) \otimes (v_1, w_1)$ as above, this relation becomes $v_1 \otimes v_1 = 0$ and $w_1 \otimes w_2 = 0$ on the first and last components. If we want the middle component to vanish as well we must ask that the isomorphism $T^1(W) \otimes T^1(V) \cong T^1(V) \otimes T^1(W)$ exchanges sign. Then we obtain that

$$\Lambda^2(V \oplus W) = \Lambda^{2,0}(V \oplus W) \oplus \Lambda^{1,1}(V \oplus W) \oplus \Lambda^{0,2}(V \oplus W)$$

by defining $\Lambda^{p,q}(V \oplus W) := \Lambda^p(V) \otimes \Lambda^q(W)$.

To make sure that this generalizes to all p, q write, e.g. $p+q=3$,

$$\begin{aligned}(v_1,w_1)\otimes(v_1,w_1)\otimes(v_1,w_1) &= (v_1,0)\otimes(v_1,0)\otimes(v_1,0)\\ &+ (v_1,0)\otimes(v_1,0)\otimes(0,w_1)\\ &+\dots\end{aligned}$$

so that imposing the relation $v\otimes v$ on $T^3(V\oplus W)$ implies imposing it on every factor. [Further arguments needed here... but looks like just boring computations...]

The main result of Hodge theory is that when we take $T^{1,0}(M)\oplus T^{0,1}(M)$, the eigenspaces of the complex structure, and pass to complex singular cohomology of a Kähler manifold we get

$$H^k(M,\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(M) \tag{1.3.1}$$

where $H^{p,q}(M)$ is the set of differential forms represented by a closed form in $\Lambda^{p,q}(M)$.

It also happens that $\overline{\Lambda^{p,q}(V)} = \Lambda^{q,p}(V)$, which is due to the fact that $\overline{\Lambda^{1,0}} = \Lambda^{0,1}$.

Exercise 1.4. Show that $(1,1)$-forms are invariant under the action of the complex structure I. And conversely, a 2-form is of type $(1,1)$ only if it is I-invariant.

Proof. Let $\alpha\wedge\beta\in\Lambda^{1,1}(M)$. Recall that $\Lambda^{1,1}(M)=\Lambda^{1,0}(M)\otimes\Lambda^{0,1}(M)$, i.e. $\alpha\in\Lambda^{1,0}(M)$ and $\beta\in\Lambda^{0,1}(M)$. For any vectors v,w tangent to some point $p\in M$,

$$\begin{aligned}I(\alpha\wedge\beta)(v,w) &= (\alpha\wedge\beta)(Iv,Iw)\\ &= \alpha(Iv)\beta(Iw)-\beta(Iv)\alpha(Iw)\\ &= -\sqrt{-1}\sqrt{-1}(\alpha(v)\beta(w)-\beta(v)\alpha(w))\\ &= \alpha(v)\beta(w)-\beta(v)\alpha(w)\\ &= (\alpha\wedge\beta)(v,w).\end{aligned}$$

The converse follows easily by the direct sum splitting of $\Lambda^2(M)$ and noting that a $(2,0)$ or $(0,2)$-form γ satisfies $I\cdot\gamma(v,w)=-\gamma(v,w)$ in an analogous computation. □

In particular this implies that ω, the fundamental form of the Riemannian metric g and the complex structure I is a $(1,1)$-form. So this shows that Kähler classes are in $H^{1,1}(M)$.
Now we shall continue discussing complex linear algebra up to Lefschetz hard theorem 1.13 and the Hodge-Riemann relations 1.15.

Definition 1.5. Let (V,g,I) be an almost complex vector space equipped with an hermitian metric g. The *Lefschetz operator* is

$$\begin{aligned}L:\Lambda^\bullet(V_{\mathbb{C}}^\vee)&\longrightarrow\Lambda^\bullet(V_{\mathbb{C}}^\vee)\\ \alpha&\longmapsto\omega\wedge\alpha.\end{aligned}$$

Recall that a metric on a vector space induces a metric on the dual vector space by defining the dual basis of an orthonormal basis to be orthonormal. And for higher exterior powers all we do is declare that wedge products of elements of the dual orthonormal basis (in ascending order) form an orthonormal basis; more precisely if $e_1,\dots,e_n$ is orthonormal for V then the elements of the form $e_{i_1}\wedge\dots\wedge e_{i_k}$ with $i_1<\dots<i_k$ are an orthonormal basis for $\Lambda^k(V^\vee)$.

Exercise 1.6. Let V be an oriented Riemannian vector space with orientation form Vol. There is a unique operator $* : \Lambda^\bullet(V^\vee) \to \Lambda^\bullet(V^\vee)$ called the *Hodge star* such that

$$\alpha \wedge *\beta = g(\alpha, \beta)\mathrm{Vol}$$

for any $\alpha, \beta \in \Lambda^k(V^\vee)$.

Proof. First prove uniqueness, then prove that mapping a basic form $e^{i_1} \wedge \ldots \wedge e^{i_k}$ to $e^{j_1} \wedge \ldots \wedge e^{j_{n-k}}$ where $(i_1, \ldots, i_k, j_1, \ldots, j_{n-k})$ is a permutation of $(1, \ldots, n)$ will work (up to sign maybe). □

Intuition on the Hodge star operator is that it maps a form to its "complement" with respect to the volume form. For instance, $*dx = dy \wedge dz$ in $\mathbb{R}^3$.

Lemma 1.7. *Hodge star is an isometry.*

Proof. Consider two basic covectors e^I and e^J. Then $*e^I$ and $*e^J$ are determined by I and J. By definition of the metric on $\Lambda^\bullet(M)$, $\left\langle e^I, e^J \right\rangle$ is zero if $I \neq J$ and 1 otherwise. Similarly, $\left\langle *e^I, *e^J \right\rangle$ is zero if $I \neq J$, while, if $I = J$, then it will clearly be 1. □

Lemma 1.8.

$$*^2 = (-1)^{k(n-k)} Id.$$

Proof. We prove that

$$\left\langle \alpha - *^2\alpha, \beta \right\rangle = 0 \qquad \forall \beta \in \Lambda^k(M).$$

$$\begin{aligned}
\langle \alpha, \beta \rangle \,\mathrm{Vol} &= \langle *\alpha, *\beta \rangle \,\mathrm{Vol} \\
&= *\beta \wedge *^2\alpha \\
&= (-1)^{\deg(*\alpha)\deg(\beta)} *^2 \alpha \wedge *\beta \\
&= \left\langle (-1)^{(n-k)k} *^2 \alpha, *\beta \right\rangle \mathrm{Vol}.
\end{aligned}$$

□

In the setting of linear algebra alone, there is no confusion on which inner product we use to define adjoint operators.

Definition 1.9. The adjoint operator of L with respect to the metric is called *dual Lefschetz operator*, i.e. it is the unique operator Λ such that

$$\langle L\alpha, \beta \rangle = \langle \alpha, \Lambda\beta \rangle \,.$$

Exercise 1.10. $\Lambda = *^{-1}\Lambda *$.

Proof.

$$\begin{aligned}
\langle L\alpha, \beta \rangle &= L\alpha \wedge *\beta \\
&= \omega \wedge \alpha \wedge *\beta \\
&= \alpha \wedge \omega \wedge *\beta \\
&= \alpha \wedge L * \beta \\
&= \alpha \wedge * *^{-1} L * \beta \\
&= \left\langle \alpha, *^{-1} L * \beta \right\rangle \mathrm{Vol}.
\end{aligned}$$

□

Next we define H, another operator on $\Lambda^\bullet$ called the *counting operator*, which is just the identity multipliied by the codegree of a form:

$$H|_{\Lambda^k V} := (k-n)\mathrm{id}.$$

Proposition 1.11. *Let $(V, \langle \cdot, \cdot \rangle)$ be an euclidean vector space endowed with an compatible almost complex structure I. Then*

$$[H, L] = 2L, \qquad [H, \Lambda] = -2\Lambda, \qquad [L, \Lambda] = H,$$

that is, there is a representation of $\mathfrak{sl}_2$ on $\Lambda^\bullet(V)$.

Definition 1.12. Let $(V, \langle \cdot, \cdot \rangle)$ be an euclidean vector space with a compatible almost complex structure I. *Primitive forms* are the elements of $\operatorname{Ker} \Lambda$. The set of primitive forms of degree k is denoted $P^k \subset \Lambda^k(V^\vee)$.

Theorem 1.13 (Lefschetz decomposition)**.**

$$\Lambda^k(V^\vee) = \bigoplus_i L^i(P^{k-2i})$$

holds and is called the Lefschetz decomposition. Moreover, the decomposition is orthogonal with respect to $\langle \cdot, \cdot \rangle$.

The upshot about this theorem is that it is proved using the fact that $\Lambda^\bullet(V^\vee)$ is an $\mathfrak{sl}_2$-representation.

Definition 1.14. Let $(V, \langle \cdot, \cdot \rangle)$ be an euclidean vector space with a compatible almost complex structure. The *Hodge-Riemann pairing* is

$$Q : \Lambda^k(V^\vee) \times \Lambda^k(^\vee) \longrightarrow \Lambda^{2n}(V^\vee) \cong \mathbb{R}$$
$$Q(\alpha, \beta) = (-1)^{\frac{k(k-1)}{2}} \alpha \wedge \beta \wedge \omega^{n-k}.$$

Theorem 1.15 (Hodge-Riemann bilinear relation)**.** *Q is positive definite on $P^{p,q}$, more precisely,*

$$\sqrt{-1}^{p-q} Q(\alpha, \bar{\alpha}) = (n-(p+q))! \, \langle \alpha, \alpha \rangle_{\mathbb{C}} > 0,$$

for $\alpha \in P^{p,q} \setminus \{0\}$ and $p + q \leq n$.

Now let's discuss the local theory on complex manifolds, which is rather just analysis on an open set of the complex numbers. We start defining $\partial, \bar{\partial}$ and aim at proving Dolbeault theorem, which is proved exactly in the same way as de Rham's theorem.

Upshot. We recall by sheaf theory that the existence of an acyclic resolution of a sheaf implies that the cohomology of the resolution equals the cohomology of the sheaf (see next paragraph). de Rham's complex is an acylic resolution of the constant sheaf $\underline{\mathbb{R}}$; acyclicity is due to Poincaré lemma - a local property. There some way of viewing the complex of singular cochains as an acyclic resolution of the constant sheaf $\underline{\mathbb{R}}$. Thus both de Rham cohomology and singular cohomology are isomorphic to the cohomology of the constant sheaf $\underline{\mathbb{R}}$.

The derived functors assign a resolution to every object in the abelian category if it has enough injectives. An acylcic resolution is such that the derived functors of all objects in the resolution vanish. The cohomology of a sheaf is defined as the resolution given by its exact functors.

Dolbeault theorem is exactly the same: Dolbeault's complex is acylcic by $\bar{\partial}$-Poincaré lemma, making it a resolution of the sheaf Ω^p.

There is an interesting relationship between the two results, which we anticipated since $\bar{\partial}$ appears to act on a "piece" of the exterior algebra, fixing another piece.

Let us start by defining ∂ and $\bar{\partial}$. Take a real-valued smooth function $f \in C^\infty(V, \mathbb{R})$. We know that the de Rham differential is given in local coordinates x^i, y^j of our $2n$ real-dimensional manifold by

$$df = \partial_{x^i} f dx^i + \partial_{y^j} f dy^j.$$

When we complexify the exterior algebra we obtain $\Lambda^\bullet_{\mathbb{C}}(V) = \Lambda^\bullet(V) \otimes_{\mathbb{R}} \mathbb{C} = \Lambda^\bullet(T^*V \otimes_{\mathbb{R}} \mathbb{C})$. Here we can multiply by complex numbers, so we can define

$$\partial_z := \frac{1}{2}(\partial_x - i\partial_y), \qquad \partial_{\bar{z}} := \frac{1}{2}(\partial_x + i\partial_y)$$

and obtain

$$dz = dx + idy, \qquad d\bar{z} = dx - idy.$$

We see that

$$\begin{aligned} \partial_z f dz + \partial_{\bar{z}} f d\bar{z} &= \frac{1}{2}(\partial_x f - i\partial_y f)(dx + idy) \\ &+ \frac{1}{2}(\partial_x f + i\partial_y f)(dx - idy) \\ &= \partial_x f dx + \partial_y f dy \\ &= df. \end{aligned}$$

Definition 1.16. Let X be an almost complex manifold. (Or vector space! The following objects may be defined over $V_{\mathbb{C}}$; we put almost complex manifold since the definition for almost complex manifolds shall be identical)

If $d : \mathcal{A}^k_{X,\mathbb{C}} \to \mathcal{A}^{k+1}_{X,\mathbb{C}}$ is the $\mathbb{C}$-linear extension of the exterior differential, then one defines

$$\partial := \Pi^{p+1,q} \circ d : \mathcal{A}^{p,q}_X \to \mathcal{A}^{p+1,q}_X, \qquad \bar{\partial} := \Pi^{p,q+1} \circ d : \mathcal{A}^{p,q}_X \to A^{p,q+1}_X$$

where $\Pi^{p,q}$ is the projection.

From this definition we see that

$$\partial f = \partial_z f dz, \qquad \bar{\partial} f = \partial_{\bar{z}} f d\bar{z}$$

for f real or complex-valued.

If we take f complex-valued and put $f = u + iv$, the condition $\bar{\partial} f = 0$ becomes the Cauchy-Riemann equations. Thus a complex-valued function is holomorphic if and only if $\bar{\partial} f = 0$, which means that its complexified de Rham differential is a $(1, 0)$-form, i.e. a multiple of dz only.

This way we see that since real partial derivatives commute,

$$\begin{aligned}\partial_z\partial_{\bar z} &= \left(\frac{1}{2}(\partial_x - i\partial_y)\right)\left(\frac{1}{2}(\partial_x + i\partial_y)\right)\\ &= \frac{1}{4}(\partial_x^2 + i\partial_x\partial_y - i\partial_y\partial_x - i^2\partial_y^2)\\ &= \frac{1}{4}(\partial_x^2 + \partial_y^2)\\ &= \frac{1}{4}(\partial_x^2 - i\partial_x\partial_y + i\partial_y\partial_x - i^2\partial_y^2)\\ &= \left(\frac{1}{2}(\partial_x + i\partial_y)\right)\left(\frac{1}{2}(\partial_x - i\partial_y)\right)\\ &= \partial_{\bar z}\partial_z,\end{aligned}$$

so that

$$\begin{aligned}\partial\bar\partial f &= \partial(\partial_{\bar z} f d\bar z)\\ &= \partial_z\partial_{\bar z} f dz \wedge d\bar z.\end{aligned}$$

And similarly,

$$\begin{aligned}\bar\partial\partial f &= \partial_{\bar z}(\partial_z f dz)\\ &= \partial_{\bar z}\partial_z f d\bar z \wedge dz\\ &= -\partial_z\partial_{\bar z} f dz \wedge d\bar z\end{aligned}$$

and thus

$$\partial\bar\partial = -\bar\partial\partial$$

on functions.

For an arbitrary (p, q)-form

$$\alpha = f_{I,J} dz^I \wedge d\bar z^J, \qquad |I| = p, |J| = q,$$

we shall also have $\partial\bar\partial = \bar\partial\partial$.

We have the following local result:

Lemma 1.17 ($\bar\partial$-Poincaré lemma on the open disc)**.** *If $\alpha \in \mathcal{A}^{p,q}(B)$ is $\bar\partial$ closed, then there exists $\beta \in \mathcal{A}^{p,q-1}$ such that $\alpha = \bar\partial\beta$.*

Proof. Looks like this is a consequence of complex analysis. □

Proposition 1.18.

$$\begin{aligned}\partial(\alpha\wedge\beta) &= \partial\alpha\wedge\beta + (-1)^{p+q}\alpha\wedge\partial\beta\\ \bar\partial(\alpha\wedge\beta) &= \partial\alpha\wedge\beta + (-1)^{p+q}\alpha\wedge\bar\partial\beta.\end{aligned}$$

Also,

$$\partial^2 = \bar\partial^2 = 0, \qquad \partial\bar\partial = -\bar\partial\partial.$$

2. Complex manifolds

Finally we arrive at complex manifolds.

Definition 2.1. A *complex manifold* M is a smooth manifold admitting an open cover $\{U_\alpha\}$ and coordinate maps $\varphi_\alpha : U_\alpha \to \mathbb{C}^n$ such that $\varphi_\alpha \circ \varphi_\beta^{-1}$ is holomorphic on $\varphi_\beta(U_\alpha \cap U_\beta) \subset \mathbb{C}^n$ for all α, β.

Example 2.2. The standard coordinate charts of projective space $\mathbb{P}^n$ are

$$U_i := \{(z_0 : \ldots : z_n) : z_i \neq 0\}$$

along with the maps

$$\begin{aligned} \varphi_i : U_i &\longrightarrow \mathbb{C}^n \\ (z_0 : \ldots : z_n) &\longmapsto \left(\frac{z_0}{z_i}, \ldots, \frac{z_{i-1}}{z_i}, \frac{z_{i+1}}{z_i}, \ldots, \frac{z_n}{z_i}\right). \end{aligned}$$

Transition functions are given by ... which are clearly holomorphic.

As a warning note that we can take the charts to be $U_i = \{z_i = 1\}$, but when considering $U_i \cap U_j$ we must be careful since this is *not* $\{z_i = 1\} \cap \{z_j = 1\}$. For example in $\mathbb{P}^1$, this would say that the intersection $U_0 \cap U_1$ is the single point $(1 : 1)$, but what about $(1 : 2)$?

Definition 2.3. An *almost complex manifold* M is a smooth manifold together with an endomorphism of the real tangent bundle $I : TM \to TM$ such that $I^2 = -\mathrm{Id}$.

Proposition 2.4. *A complex manifold has a natural almost complex structure.*

Proof. Define bundle maps I_i on local trivializations U_i by rotating vectors 90 degrees in counter-clockwise direction at every copy of $\mathbb{C}$ in the tangent space $\mathbb{C}^n$. The upshot is that since X is a complex manifold we can interpret the I_i as multiplication by $\sqrt{-1}$ in the complex vector space structure of the tangent space.

Then the changes of coordinates satisfy the condition needed to glue to a global endomorphism, namely $I_j = g_{ij} I_i g_{ij}^{-1}$, trivially because the g_{ij} are holomorphic and thus $\mathbb{C}$-linear.

Addendum. It looks like we can just directly define the I_i by multiplication by $\sqrt{-1}$. □

But what about the converse? What's the condition for an almost complex manifold to be complex?

Definition 2.5. An almost complex structure on a smooth manifold X is called *integrable* if it is induced from a complex structure on X as in Proposition 2.4.

Theorem 2.6 (Newlander-Nirenberg)**.** *An almost complex structure I on a smooth manifold is integrable if and only if $[T^{0,1}, T^{0,1}] \subset T^{0,1}$.*

Other equivalent conditions...

The *tensor* is given by

$$N_I[X, Y] = [X, Y] + I[IX, Y] + I[X, IY] - [IX, IY]$$

Now let's discuss Dolbeault cohomology.

Proposition 2.7. *The space $H^0(X, \Omega^p_X)$ of holomorphic p-forms on a complex manifold is the subspace $\{\alpha \in \mathcal{A}^{p,0}(X) : \bar{\partial} = 0\}$.*

Proof. The problem is local. We use that $f dz_{i_1} \wedge \ldots \wedge dz_{i+p}$ is holomorphic if and only if f is holomorphic, which is equivalent to

$$\bar{\partial}(f dz_{i_1} \wedge \ldots \wedge dz_{i_p} = \sum \frac{\partial f}{\partial \overline{z}_i} d\overline{z}_i \wedge dz_{i_1} \wedge \ldots \wedge dz_{i_p} = 0.$$

□

For a more general statement we must define the (p,q)-*Dolbeault cohomology* by

$$H^{p,q}(X)_{\bar{\partial}} := \frac{\mathrm{Ker}(\bar{\partial} : \mathcal{A}^{p,q}(X) \to \mathcal{A}^{p,q+1}(X)}{\mathrm{Im}(\bar{\partial} : \mathcal{A}^{p,q-1}(X) \to \mathcal{A}^{p,q}(X))}.$$

Then

Theorem 2.8 (Dolbeault)**.** *The Dolbeault cohomology of X computes the cohomology of the sheaf Ω_X^p, i.e. $H_{\bar{\partial}}^{p,q}(X) = H^q(X, \Omega_X^p)$.*

3. Kähler manifolds, Kähler identities

Definition 3.1. A *hermitian structure* on a complex manifold M is a Riemannian metric g that is also an isometry for the complex structure I, that is $g(I\cdot, I\cdot) = g(\cdot,\cdot)$.

Definition 3.2. If (M, I, g) is a hermitian manifold, $\omega(\cdot,\cdot) := g(I\cdot,\cdot)$ is called the *fundamental form.* (M, I, g) is called *Kähler* if its fundamental form is closed.

As in with vector spaces endowed with almost complex structures, we have the operators $L, \Lambda, *$ defined on the cotangent spaces of a complex manifold. As a corollary we have that Lefschetz decomposition passes to exterior algebra of a hermitian manifold:

Lemma 3.3. *Let (X, g) be an hermitian manifold. Then there exists a direct sum decomposition of vector bundles*

$$\Lambda^k(X) = \bigoplus_{i \geq 0} L^i(P^{k-2i}X),$$

there $P^{k-2i}X$ is the kernel of Λ.

Apart from the linear-algebraic operators, on manifolds we have other operators.

Definition 3.4. The *adjoint de Rham differential* is

$$d^* := (-1)^{n(k+1)+1} * \circ d \circ * : \mathcal{A}^k(M) \to \mathcal{A}^{k-1}(M)$$

where, as always, $\mathcal{A}^k(M)$ denotes the sections of $\Lambda^k(M) := \Lambda^k(T^*M)$.

Definition 3.5. The *Laplace operator* is

$$\Delta = dd^* + d^*d.$$

If M is a complex manifold, then

$$d^* = - * \circ d \circ *.$$

Similarly we define

$$\partial^* := - * \circ \partial \circ *, \qquad \bar{\partial}^* := - * \circ \bar{\partial} \circ *,$$

which yield the Laplacians

$$\Delta_\partial = \partial\partial^* + \partial^*\partial,$$
$$\Delta_{\bar\partial} = \bar\partial\bar\partial^* + \bar\partial^*\bar\partial.$$

Proposition 3.6 (Kähler identities)**.** *Let X be a complex manifold endowed with a Kähler metric g. Then*

1. $[\bar\partial, L] = [\partial, L] = 0$*,* $[\bar\partial^*, \Lambda] = [\partial^*, \Lambda] = 0$.
2. *Etc.*
3. $\Delta_\partial = \Delta_{\bar\partial} = \frac{1}{2}\Delta$.
4. Δ *commutes with* $*, \partial, \bar\partial, \partial^*, \bar\partial^*, L$ *and* Λ.

Definition 3.7. On a complex manifold (M, I),

$$d^c := I^{-1}dI = -IdI.$$

Notice (prove) that

$$d^c = \frac{i}{2}(\bar\partial - \partial)$$

so that

$$dd^c = i(\partial\bar\partial).$$

We may also define

$$d^{c*} := - * \circ d^c \circ *.$$

Now let's introduce Fubini-Study form. The following discussion of psh functions is because to pass from a potential, i.e. a function, to a $(1,1)$-form we apply dd^c (or $\partial\bar\partial$). Now if the potential is psh, then the Hessian will be positive definite, which is just the same as saying that $dd^c f$ is positive definite $(1,1)$-form.

Definition 3.8. Let V be a real vector space equipped with a complex structure I. A $(1,1)$-form ω is *positive* if $\omega(x, Ix) \geq 0$ for any nonzero $x \in V$. (Notice the I is in the second entry!) It is called *Hermitian* or *semipositive* if $\omega(x, Ix) > 0$ for all nonzero $x \in V$.

See [Huy05, Lemma 3.1.7] to find Huybrecht's definition of a closed *positive definite* $(1,1)$-form, namely if $\omega = \frac{i}{2}\sum h_{ij} dz_i \wedge d\bar z_j$ such that $(h_{ij}(x))$ is a positive definite hermitian matrix for any $x \in X$.

The point is that this condition is the same as saying that the associated Riemannian metric is in fact positive definite. Indeed, $\omega(\cdot, I\cdot) = g(\cdot,\cdot)$ which is positive definite.

Finally observe that numerical positivity of the Kähler class, i.e. positivity with respect to the intersection form, follows exactly from this fact: we shall have that $\int_Y \omega^p > 0$ for any analytic cycle Y.

Definition 3.9. A function f on a complex manifold is *plurisubharmonic* if $dd^c f$ is a positive $(1,1)$-form.

Again: psh is equivalent to the Hermitian matrix

$$\left(\frac{\partial^2 f}{\partial z_i \partial \bar z_j}\right)$$

being positive semidefinite. ChatGPT: it is also equivalent to: its restriction to any complex line being subharmonic ($\Delta f \geq 0$).

Exercise 3.10. Let $z_1, \dots, z_n$ be the standard coordinates on $\mathbb{C}^n$, and $l(z_1, \dots, z_n) := \sum |z_i|^2$. Prove that $dd^c \log l = \frac{dd^c l}{l} - \frac{dl \wedge d^c l}{l^2}$ on $\Omega = \mathbb{C}^n \setminus 0$. Prove that the Hermitian form $dd^c \log l$ is degenerate and can be written as $(-\sqrt{-1}) \sum_{i=1}^{n-1} \alpha_i \xi_i \wedge \bar{\xi}_i$, where ξ_i is a local frame in $\Lambda^{1,0}(\Omega)$ and α_i are positive functions.

Proof. For the first equation we just use the fact (why?) that both d^c and d satisfy the chain rule. For the second equation I computed that

$$dd^c l = \sqrt{-1} \sum dz_i \wedge d\bar{z}_i. \tag{3.10.1}$$

But I also need to compute $d^c l = \frac{\sqrt{-1}}{2}(\bar{\partial} - \partial) l$ and $dl = (\partial + \bar{\partial}) l$. I find that $dl \wedge d^c l = \sqrt{-1} \partial l \wedge \bar{\partial} l$. But I also found that $\partial l = \sum \bar{z}_i dz_i$ and $\bar{\partial} l = \sum z_i d\bar{z}_i$. So I get

$$dl \wedge d^c l = \sqrt{-1} \left(\sum \bar{z}_i dz_i \right) \wedge \left(\sum z_i d\bar{z}_i \right) = \sqrt{-1} \sum \bar{z}_i z_j dz_i \wedge d\bar{z}_j \tag{3.10.2}$$

Now we join Equations 3.10.1 and 3.10.2 to find the coefficient functions with respect to $dz_i \wedge d\bar{z}_i$. From Equation 3.10.1 we get coefficient functions δ_{ij}, and from Equation 3.10.2 we get $\bar{z}_i z_j$ so

$$dd^c \log l = \sqrt{-1} \sum \left(\frac{\delta_{ij}}{l} - \frac{\bar{z}_i z_j}{l^2} \right) dz_i \wedge d\bar{z}_j.$$

I'm not sure how to prove that the coefficient functions are positive... Or that the form is degenerate... But fortunately this can be checked at [Huy05, p. 118]. □

The point of this exercise is to show that the form $dd^c \ell$ is almost an Hermitian form — but it has a kernel. I think the point of the next exercises in this handout is to show that the kernel is actually the "radial directions". And the point of that is to take quotient by $\mathbb{C}^*$ action, which is literally passing to the projective space.

A bit more formally:

Definition 3.11. Let M be a manifold and $\omega \in \Omega^k(M)$. Suppose that $\pi : M \to B$ is a surjective submersion with connected fibers. We say that ω is *basic* (with respect to π) is there exists a form $\overline{\omega} \in \Omega^k(B)$ such that $\pi^* \overline{\omega}$.

Exercise 3.12.
1. Show that ω is basic if and only if $i_X \omega = 0$ and $L_X \omega = 0$ for all vector fields X tangent to the fibers of π. In particular, if ω is closed, show that it is basic if $\mathrm{Ker}(\pi_*) \subseteq \mathrm{Ker}(\omega)$ (pointwise in M).
2. Suppose that ω is a closed 2-form on M and $\mathrm{Ker}(\pi_*) = \mathrm{Ker}(\omega)$. Show that $\omega = \pi^* \overline{\omega}$ and $\overline{\omega} \in \Omega^2(B)$ is symplectic.
3. (Application to reduction.)

Proof.
(1) Since π is surjective, for any vector $\bar{X}$ tangent to B there is a vector X tangent to M such that $\pi_* X = \bar{X}$. Define a form $\overline{\omega}(\bar{X}^1, \dots, \bar{X}^k) := \omega(X^1, \dots, X^k)$.

To show this does not depend on the choice of lift of $\bar{X}$ just notice that the tangent space of M can be written $T_p M \simeq T_p F_b \oplus \pi^*(T_b B)$ where $\pi(p) = b$. Since ω kills vectors tangent to the fibers, $\overline{\omega}$ is well-defined.

Notice that there can be different choices of p such that $\pi(p) = b$. We must show that the definition is also independent of this choice. Let $X \in T_p M$ and $Y \in T_q M$ be such that $\pi(p) = \pi(q) = b \in B$ and $\pi_* X = \pi_* Y$. We wish to show that $\omega(\dots, X, \dots) = \omega(\dots, Y, \dots)$.

This a little more involved so I'll just sketch it for now. If there is a vector field V tangent to the fibers whose flow φ joins p and q, we are done, since we pull back ω along this flow, and by the condition of the Lie derivative, i.e. $L_V\omega$ for V tangent to the fiber, we see that $\varphi^*\omega = \omega$. (Indeed, $L_V\omega = \frac{d}{dt}\varphi^*\omega$, so the derivative is zero and at $t = 0$ we have ω.)

Note that the choice of Y within T_qM is superfluous since we already know that ω is constant within every tangent space at a poi

...

□

Exercise 3.13. Let X be a connected complex manifold of dimension $n > 1$ and let g be a Kähler metric. Show that g is the only Kähler metric in its conformal class, i.e. if $g' = e^f g$ is Kähler then f is constant.

Exercise 3.14. Let X be a connected complex manifold and h be a Kähler metric. Let ω be the associated Kähler form. Show that if $\dim X \geq 2$, the wedge product with ω is injective on 1-forms.

Show that if $\dim X \geq 2$ and ϕ is a differentiable function with values in $\mathbb{R}^+$ such that ϕh is also a Kähler metric, then ϕ is constant.

Proof. Take local symplectic coordinates, i.e. such that $\omega = \sum_i dz^i \wedge d\bar{z}^i$. Let α be any 1-form and suppose it is written in those coordinates as $\alpha = \sum_i a_i dz^i + \sum_i \bar{a}_i d\bar{z}^i$. Then

$$\begin{aligned}\alpha \wedge \omega &= \left(\sum_i a_i dz^i\right) \wedge \left(\sum_i dz^i \wedge d\bar{z}^i\right) + \left(\sum_i \bar{a}_i dz^i\right) \wedge \left(\sum_i dz^i \wedge d\bar{z}^i\right) \\ &= \sum_{i,j} a_i dz^i \wedge dz^j \wedge d\bar{z}^j + \sum_{i,j} \bar{a}_i d\bar{z}^i \wedge dz^j \wedge d\bar{z}^j.\end{aligned}$$

We see that if $\alpha \wedge \omega = 0$ then both its $1,2$ and $2,1$ parts are zero, which foces a_i and $\bar{a}_i$ to vanish, i.e. $\alpha = 0$. □

Exercise 3.15. Let E be a holomorphic vector bundle of rank r over a complex manifold X. Show that if L is a holomorphic line bundle on X, then $\mathbb{P}(E^* \otimes L^*) = \mathbb{P}(E^*)$ but the line bundle $\mathcal{O}_{\mathbb{P}(E^*\otimes L^*)}(1)$ is isomorphic to $\mathcal{O}_{\mathbb{P}(E^*)}(1) \otimes \pi^* L$, where $\pi : \mathbb{P}(E^*) \to X$ is the structural morphism.

Exercise 3.16. (M, I, g) is Kähler if and only if $\nabla^{LC} I = 0$.

4. Hodge theory

On a compact hermitian manifold (X, g) we may define

$$(\alpha, \beta) := \int_X g_{\mathbb{C}}(\alpha, \beta) * 1,$$

which is an hermitian product on $\mathcal{A}^*_{\mathbb{C}}(X)$.

Lemma 4.1. *Let X be a compact hermitian manifold. Then ∂^* and $\bar{\partial}^*$ are adjoints to ∂ and $\bar{\partial}$ with respect to $(\cdot, \cdot)$.*

Proof. We have

$$(\partial\alpha,\beta) = \int_X \langle \partial\alpha,\beta\rangle_{\mathbb{C}} \operatorname{Vol} = \int_X \partial\alpha \wedge *\beta,$$
$$(\alpha,\partial^*\beta) = \int_X \langle \alpha, -*\partial*\beta\rangle_{\mathbb{C}} \operatorname{Vol} = \int_X \alpha\wedge\partial*\beta.$$

Now we also have $\partial(\alpha\wedge\beta) = \partial\alpha\wedge\beta + \alpha\wedge\partial\beta$, and the integral of the first vanishes by Stokes theorem. □

Lemma 4.2. *Let X be a compact hermitian manifold. A form is $\bar\partial$-harmonic if and only if it is $\bar\partial$-closed and $\bar\partial^*$ closed. The same holds for ∂, and in fact, on any compact oriented Riemannian manifold (M,g), a similar result holds for the usual d-Laplacian.*

Proof.

$$\begin{aligned}(\Delta\alpha,\alpha) &= ((dd^*+d^*d)\alpha,\alpha)\\ &= (dd^*\alpha,\alpha) + (d^*d\alpha,\alpha)\\ &= (d^*\alpha,d^*\alpha) + (d\alpha,d\alpha)\\ &= \|d^*\alpha\|^2 + \|d\alpha\|^2.\end{aligned}$$

□

The following is the first of the symmetries of the Hodge diamond:

Lemma 4.3. *For any Riemannian manifold (M,g), not necessarily compact, $*\Delta = \Delta*$ and $*: \mathcal{H}^k(M,g) \cong \mathcal{H}^{m-k}(M,g)$.*

In the hermitian case, we have for a compact hermitian connected manifold (M,g), that the pairing

$$\begin{aligned}\mathcal{H}^{p,q}_{\bar\partial} \times \mathcal{H}^{n-p,n-q}_{\bar\partial} &\longrightarrow \mathbb{C}\\ (\alpha,\beta) &\longmapsto \int_X \alpha\wedge\beta\end{aligned}$$

is non-degenerate and thus

$$\mathcal{H}^{p,q}_{\bar\partial}(X,g) \cong \mathcal{H}^{n-p,n-q}_{\bar\partial}(X,g),$$

which is called *Serre duality*, the second of the symmetries of the Hodge diamond. We shall later generalize this Serre duality to other vector bundles.

There is a third symmetry of the Hodge diamond, given by the Lefschetz operator. Since $[L,\Delta]=0$, one of the Kähler identities, we see that L maps harmonic forms to harmonic forms. Further, since $L^{n-k}: \Lambda^k(V^\vee) \to \Lambda^{2n-k}(V^\vee)$ is bijective we have that the induced map

$$L^{n-k}: \mathcal{H}^{p,k-p}(X,g) \to \mathcal{H}^{n+p-k,n-p}(X,g)$$

is injective. It is also bijective since… thus it is an isomorphism.

Now we are almost ready for Hodge theorem.

Proposition 4.4. *Let (X,g) be a compact hermitian manifold. The degree, bidegree and Lefschetz decomspositions are orthogonal with respect to $(\cdot,\cdot)$.*

The bidegree decomposition passes to ∂ and $\bar\partial$-harmonic forms on any compact hermitian manifold, but only for Kähler manifolds does it pass to d-harmonic forms.

Proposition 4.5. *Let (X,g) be a hermitian manifold, not necessarily compact.*

1. $\mathcal{H}^k_{\bar\partial}(X,g) = \bigoplus_{p+q=k} \mathcal{H}^{p,q}_{\bar\partial}(X,g)$ *and* $\mathcal{H}^k_{\partial}(X,g) = \bigoplus_{p+q=k} \mathcal{H}^{p,q}_{\partial}(X,g)$.
2. *If (X,g) is Kähler then both decompositions coincide with* $\mathcal{H}^k(X,g) = \bigoplus_{p+q=k} \mathcal{H}^{p,q}(X,g)$. *In particular,* $\mathcal{H}^k(X,g)_{\mathbb{C}} = \mathcal{H}^k_{\bar\partial}(X,g) = \mathcal{H}^k_{\partial}(X,g)$.

We finally arrive at the Hodge decomposition. The first result concerns only compact Riemannian manifolds, and "requires some hard, but by now standard, analysis":

Theorem 4.6 (Hodge decomposition)**.** *Let (M,g) be a compact Riemannian manifold. Then*

$$\mathcal{A}^k(M) = d(\mathcal{A}^{k-1}(M)) \oplus \mathcal{H}^k(M,g) \oplus d^*(\mathcal{A}^{k+1}(M))$$

which is orthogonal with respect to $(\cdot,\cdot)$. Moreover, the space $\mathcal{H}^k(M)$ is finite-dimensional.

And now we also have

Theorem 4.7 (Hodge decomposition)**.** *Let (X,g) be a compact hermitian manifold. Then*

$$\mathcal{A}^{p,q}(X) = \partial\mathcal{A}^{p-1,q}(X) \oplus \mathcal{H}^{p,q}_{\partial}(X,g) \oplus \partial^*\mathcal{A}^{p+1,q}(X)$$

and

$$\mathcal{A}^{p,q}(X) = \bar\partial\mathcal{A}^{p-1,q}(X) \oplus \mathcal{H}^{p,q}_{\bar\partial}(X,g) \oplus \bar\partial^*\mathcal{A}^{p+1,q}(X).$$

If (X,g) is Kähler, then $\mathcal{H}^{p,q}_{\partial}(X,g) = \mathcal{H}^{p,q}_{\bar\partial}(X,g)$.

Upshot: if X is Kähler we can pass from usual harmonic (p,q)-forms to $\bar\partial$-harmonic (p,q)-forms, which takes us to $H^{p,q}(X)$. To pass from $H^k(X)$ to $\mathcal{H}^k(X)$ and from $\mathcal{H}^{p,q}_{\bar\partial}(X)$ to $H^{p,q}(X)$ we have the following two lemmas which in fact are corollaries from the corresponding decompositions (Theorems 4.6 and 4.7, respectively).

Lemma 4.8. *Let (M,g) be a compact oriented Riemannian manifold. The natural map $\mathcal{H}^k(M) \to H^k(M,\mathbb{R})$ that associates to any harmonic form its cohomology class is an isomorphism.*

Lemma 4.9. *Let (X,g) be a compact hermitian manifold. Then the canonical projection $\mathcal{H}^{p,q}_{\bar\partial}(X,g) \to H^{p,q}(X)$ is an isomorphism.*

Before we pass to Hodge theorem we state one final key result

Lemma 4.10 (ddbar Lemma)**.** *Let X be a compact Kähler manifold. If α is a d-closed (p,q)-form, it is equivalent that α is*

1. *d-exact,*
2. *∂-exact,*
3. *$\bar\partial$-exact,*
4. *$\partial\bar\partial$-exact.*

There is another version in terms of $dd^c := I^{-1}dI = \frac{1}{2}(\partial - \bar\partial)$.

Lemma 4.11 (ddc Lemma)**.** *Let X be a compact Kähler manifold. Any d-closed and d^c-exact form form is dd^c-exact.*

That's all the ingredients for a (very rough) idea on why the Hodge theorem holds. Upshot:

$$H^k(X,\mathbb{C}) = \mathcal{H}^k(X) = \bigoplus_{p+q=k} \mathcal{H}^{p,q}(X) = \bigoplus_{p+q=k} \mathcal{H}^{p,q}_{\bar\partial}(X) = \bigoplus_{p+q=k} H^{p,q}(X).$$

(1) Start with singular complex cohomology,
(2) pass to harmonic forms via Lemma 4.8 (this is where heavy real analysis enters; what Huybrechts calls Hodge decomposition),
(3) decompose harmonic forms in (p,q)-types via Proposition 4.5 (using Kählerness),
(4) use that $\mathcal{H}^{p,q} \cong \mathcal{H}^{p,q}_{\bar\partial}$ by the same Proposition 4.5 (using Kählerness),
(5) go back to Dolbeault cohomology via Lemma 4.9 (again Hodge decomposition, now for $\bar\partial$).

Theorem 4.12 (Hodge)**.** *Let (X,g) be a compact Kähler manifold. Then there exists a decomposition*

$$H^k(X,\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X)$$

which does not depend on the Kähler structure. Moreover, with respect to complex conjugation on $H^\bullet(X,\mathbb{C}) = H^\bullet(X,\mathbb{R})\otimes\mathbb{C}$ we have $\overline{H^{p,q}(X)} = H^{q,p}(X)$, and, by Serre duality on harmonic forms, $H^{p,q}(X) \cong H^{n-p,n-q}(X)^\vee$.

Exercise 4.13. Show that $\mathrm{Pic}(\mathbb{P}^n) \simeq \mathbb{Z}$.

The next step is to check that most results we've developed so far for complex vector spaces and complex manifolds descend to cohomology in Kähler manifolds.

Let's put here the part on the cones until I find a better place.

Recall that $\mathrm{NS}(X)$ is the quotient group $\mathrm{Pic}(X)/H^1(\mathcal{O}_X) \subset H^2(X,\mathbb{Z}) \subset H^2(X,\mathbb{R})$ coming from the exponential exact sequence. This has an intersection form. In the case of a K3 we readily see that $h^1 = 0$ so $\mathrm{NS} = \mathrm{Pic}$. In fact, the intersection form coincides with the usual cup product on 2-cocycles of singular cohomology.

For a Kähler manifold we have

- The *Kähler cone*. Classes in $H^{1,1}(X,\mathbb{R})$ that are represented by a Kähler form.
- The *positive cone*. Classes $\mathrm{NS}(X)_\mathbb{R} := \mathrm{NS}(X)\otimes_\mathbb{Z}\mathbb{R}$ with $\alpha^2 > 0$.
- The *ample cone*. Classes $\mathrm{NS}(X)_\mathbb{R}$ with $\alpha^2 > 0$ and $\alpha.C > 0$ for all algebraic curves 0. (See 16.1.) This is true, but hard to prove, see [**?**, p. 150]. Another description is that the ample cone is the set of classes in $\mathrm{NS}(X)_\mathbb{R}$ that are finite sums $\sum a_i L_i$ for $L_i \in \mathrm{NS}(X)$ ample and $a_i \in \mathbb{R}_{>0}$.
- The *nef cone*. Classes (in $\mathrm{NS}(X)$?) with $\alpha.C = \int_C \alpha \geq 0$ for any algebraic curve. There is also a notion of nef for analytic cycles which basically says that the form is approximated by Kähler forms. In [DP01, Theorem 4.7(ii)] we have that a $(1,1)$-class is nef (in the analytic sense) if and only if $\int_Y \alpha\wedge\omega^{p-1} \geq 0$ for every irreducible analytic set $Y \subset X$, $p = \dim Y$ and $\bar\omega$ Kähler class.

The upshot about Demailly-Paŭn theorem is that Kähler classes are numerically positive (recall that "numerically" means w.r.t. intersection form) on *analytic* cycles. This distinguishes complex analytic geometry from algebraic geometry. This is why Demailly-Paŭn say that their theorem is a generalization of Nakai-Moishezon: because the latter is for *algebraic* cycles, indeed, found in Hartshorne's book.

The proof goes roughly as follows. It's clear that $\mathcal{K} \subset \mathcal{P}$ ($\mathcal{P}$ is the analytically numerically positive cone) because, since the Kähler form comes from the Riemannian metric and the curves are complex, which means that they are invariant under the complex structure, we have integrals $\int_Y \omega$ for curves $Y \subset X$ but $\omega(v, Jv) > 0$ for a basis v, Jv of $T_p Y$. That is to say, there are no complex Lagrangian curves.

But that's elementary. The point of the theorem is proving that a form in $\overline{\mathcal{K}} \cap \mathcal{P}$, i.e. in the closure of the Kähler cone and which is also numerically analytically positive, is in fact Kähler.

So, the theorem says that the boundary of the Kähler cone does not intersect the numerically analytically positive cone.

Theorem 4.14 (Demailly-Paŭn)**.** *Let X be a compact Kähler manifold. The Kähler cone is a connected component of the cone of 1,1-classes numerically positive on analytic cycles.*

Exercise 4.15. Let M be a K3 surface such that $\mathrm{rkNS}(M) < 20$. Using Demailly-Paun theorem, prove that there exists a non-zero vector $\eta \in H^{1,1}(M)$ which satisfies $\eta^2 = 0$ and belongs to the boundary of the Kähler cone.

Proof. Let $\alpha \in \mathrm{NS}(X)^\perp \setminus \{0\}$, which is nonempty since $\dim \mathrm{NS}(X) < 20 = h^{1,1}$. By the Hodge index theorem 5.2 the intersection form on $H^{1,1}(X, \mathbb{R})$ has index $(1, 19)$. If we assume that X is projective then we know that the restriction of the Fubini-Study form ω is integral. Since ω is positive (by being a Kähler form), we see by the index of the intersection form that α must be negative.

Define $\eta_t := t\alpha + (1-t)\omega$ for $t \in [0, 1]$. Then $\eta_t^2 = t^2\alpha^2 + (1-t)^2\omega^2$. Since at $t = 0$ we have a positive number and at $t = 1$ a negative number, we know that there exists t_0 such that $\eta_{t_0}^2 = 0$ by continuity of the intersection form.

Now we need to show that η_{t_0} is in the boundary of the Kähler cone. For this it's enough to show that for every $\varepsilon > 0$ $\eta_{t_0+\varepsilon}$ is in the Kähler, which by Demailly-Păun means that we have to show that it is positive on analytic cycles.

By our choice of ω and α we see that

$$\int_Y \eta_{t_0+\varepsilon} = \int_Y (t_0 + \varepsilon)\alpha + (1 - (t_0 + \varepsilon))\omega$$

and the integral of α vanishes since α is in the orthogonal complement of $\mathrm{NS}(X)$, which contains all analytic curves by X being projective (Chow theorem). □

5. Lefschetz theorems

Theorem 5.1 (Lefschetz theorem on (1,1)-classes)**.** *All the classes in $H^{1,1}(X, \mathbb{Z}) := H^{1,1}(X) \cap H^2(X, \mathbb{Z})$ are algebraic.*

This implies that $\mathrm{NS}(X) := \mathrm{Pic}(X)/\ker c_1$ is isomorphic to $H^{1,1}(X, \mathbb{Z})$. The "realification" of the integer classes, $H^{1,1}(X, \mathbb{Z}) \otimes \mathbb{R} = \mathrm{NS}(X) \otimes \mathbb{R}$ is a vector subspace $H^{1,1}(X) \cap H^2(X, \mathbb{R})$.

Theorem 5.2 (Hodge index)**.** *The index of the intersection pairing on $H^{1,1}(X, \mathbb{R})$ is $(1, \rho(X) - 1)$.*

Theorem 5.3 (Hard Lefschetz Theorem)**.** *The map*

$$L^k : H^{n-k}(M) \to H^{n+k}(M)$$

is an isomorphism. If we define the primitive cohomology as

$$P^{n-k}(M) = KerL^{k+1}$$

we have

$$H^m(M) = \bigoplus_k L^k P^{m-2k}(M),$$

called the Lefschetz decomposition.

6. Vector bundles, connections, curvature and Chern classes

Lemma 6.1 (Vector bundle construction from fibers)**.** *Suppose M is a complex manifold, and for each $p \in M$ we are given a k-dimensional vector space E_p. Let $E = \coprod_{p \in M} E_p$ and let $\pi : E \to M$ be the obvious projection. Suppose further that we are given*

1. *an indexed open cover U_i of M*
2. *for each i, a bijection $\psi_i : \pi^{-1}(U_i) \to U_i \times \mathbb{C}^k$,*
3. *for each i, j with $U_i \cap U_j \neq \emptyset$ a holomorphic map $g_{ij} : U_i \cap U_J \to GL(k, \mathbb{C})$ such that $\psi_i \circ \psi_j^{-1}(p, v) = (p, g_{ij}(p)v$.*

Then $\pi : E \to M$ has a unique structure as a holomorphic vector bundle with (U_i, ψ_i) as holomorphic local trivializations, and with g_{ij} as transition functions.

Proposition 6.2 (Vector bundle construction from cocycle)**.** *Let M be a complex manifold and let U_i be an open cover of M. Suppose that for each i, j such that $U_i \cap U_j \neq \emptyset$ we are given a holomorphic map $g_{ij} : U_i \cap U_j \to GL(k, \mathbb{C})$ such that*

$$g_{ij}(p)g_{jk}(p)g_{ki}(p) = Id$$

for any $p \in U_i \cap U_j \cap U_k$.

Then there is a rank-k holomorphic vector bundle $E \to M$ that admits a holomorphic trivialization over each set U_i with transition functions given by g_{ij}.

Proposition 6.3 (Holomorphic Vector Bundle Isomorphism Criterion)**.** *Two vector bundles E and F with transition functions g_{ij} and h_{ij} are isomorphic if and only if $g_{ij} = A_i^{-1} h_{ij} A_j$ for some $A_i : U_i \to GL_n(\mathbb{C})$.*

Proof. The idea is to define a map in local trivializations and confirm that this glues to a global bundle map. This means: invent $\varphi_i : E(U_i) := \Gamma_E(U_i) \to F(U_i)$, i.e. a local map of sections (yes, a map of sections, since a bundle map is constructed as map of the total spaces, which are points and vectors, but we need to preserve the projection, so all we care is about the vectors). Then check that on $U_i \cap U_j := U_{ij}$ we have $\varphi_i = \varphi_j$. This literally creates a bundle map.

Try to do it for line bundles, so A_i are just functions. Take a local frame e_i for E and a local frame f_j for F. Change coordinates and apply the map, apply the map and change coordinates - this should give both sides of the cocycle condition of the proposition. □

Exercise 6.4 (Tensor product bundle)**.** Let E and F be vector bundles determined by the cocycles $g_{ij} : U_i \cap U_j \to \mathrm{GL}(r, \mathbb{C})$ and $\tilde{g}_{ij} : U_i \cap U_j \to \mathrm{GL}(\tilde{r}, \mathbb{C})$. Verify the cocycle description for

- The direct sum $E \oplus F$,
- the tensor product $E \otimes F$ and
- the dual bundle $E^\vee$.

Proof. Let's do the tensor product. I think that, slightly more generally than in the formulation of the question, if U_i is a trivializing cover for E and $\tilde{U}_j$ is a trivializing cover for F, we should consider the cover $U_i \cap \tilde{U}_j$. Let us denote such a cover simply by U_i.

Define a cocycle by

$$\begin{aligned} h_{ij} : U_i \cap U_j &\longrightarrow \mathrm{GL}(r\tilde{r}, \mathbb{C}) \\ p &\longmapsto g_{ij}(p) \otimes \tilde{g}_{ij}(p). \end{aligned}$$

The key observation is that while formally the image of h_{ij} lies in $\mathrm{GL}(\mathbb{C}^r \otimes \mathbb{C}^{\tilde{r}})$, we can regard such a space as $\mathrm{GL}(r\tilde{r}, \mathbb{C})$ by defining, for $A \in \mathrm{GL}(r, \mathbb{C})$ and $A' \in \mathrm{GL}(\tilde{r}, \mathbb{C})$, the map $A(v^i e_i) \otimes \tilde{A}(\tilde{v}^j \tilde{e}_j) = v^i \tilde{v}^j A e_i \otimes \tilde{A} \tilde{e}_j$ where e_i and $\tilde{e}_j$ are bases of $\mathbb{C}^r$ and $\mathbb{C}^{\tilde{r}}$. This is indeed a linear transformation of the $r\tilde{r}$-dimensional $\mathbb{C}$-vector space generated by the basis $e_i \otimes \tilde{e}_j$.

Thus the cocycle condition follows simply by the cocycles of E and F, and commutativity of the coordinate functions. □

Example 6.5 (Determinant bundle)**.** Let $\pi : E \to M$ be a vector bundle of rank r. The *determinant bundle* of E is given by the following data. A space $\tilde{E} := \coprod_{p \in M} \Lambda^r E_p$ with the obvious projection map and topology induced from such map. Since we have a cover U_i of M along with local trivializations $U_i \overset{\varphi_i}{\simeq} U_i \times \mathbb{C}^r$ we may just postcompose φ_i with the functor Λ^r on the second entry, giving trivializations

$$\pi(U_i) \xrightarrow[\simeq]{\varphi_i} U_i \times \mathbb{C}^r \xrightarrow{\mathrm{Id} \times \Lambda^r} U_i \times \Lambda^r \mathbb{C}^r = U_i \times \mathbb{C}$$

We can also apply Λ^r to the transition functions g_{ij} on the intersections $U_{ij} := U_i \cap U_j$

$$\begin{tikzcd} U_{ij} \arrow[r, "\varphi_i"] & U_{ij} \times \mathbb{C}^r \arrow[d, "g_{ij} \in \mathrm{GL}_r(\mathbb{C})"] \\ U_{ij} \arrow[r, "\varphi_j"] & U_{ij} \times \mathbb{C}^r \end{tikzcd}$$

giving maps in $\mathrm{GL}_1(\mathbb{C})$. Recall that Λ^r is functorial on linear maps $g_{ij} \in \mathrm{GL}_n(\mathbb{C})$ because it is given by the determinant of g, and the determinant is multiplicative. We thus obtain the cocycle condition on the maps g_{ij} after applying Λ^r.

Example 6.6 (Tautological bundle)**.** It's the bundle denoted $\mathcal{O}_{\mathbb{P}^n}(-1)$ whose fiber over a point is the line that the point is. To show this is a line bundle take the canonical trivializing cover from Example 2.2. A local trivialization of $\mathcal{O}(-1)$ is given by

$$\begin{aligned} \psi_i : \pi^{-1}(U_i) &\longrightarrow \varphi_i(U_i) \times \mathbb{C} \\ p = (z_0 : \ldots : z_n) &\longmapsto (\varphi(p), \lambda) \end{aligned}$$

where the standard coordinate charts φ_i are defined by

$$\varphi_i(p) := \left(\frac{z_0}{z_i}, \ldots, \frac{z_{i-1}}{z_i}, \frac{z_{i+1}}{z_i}, \ldots, \frac{z_n}{z_i} \right)$$

and λ is just the coordinate of a vector

$$\lambda \left(\frac{z_0}{z_i}, \ldots, \frac{z_{i-1}}{z_i}, 1, \frac{z_{i+1}}{z_i}, \ldots, \frac{z_n}{z_i} \right).$$

This makes sense since what we want is that the fiber of $\mathcal{O}(-1)$ is the 1-dimensional space spanned by the point p seen in $\mathbb{C}^{n+1}$. Also notice that π is the projection from the total space, which is implicitly defined as the disjoint union of the $\varphi_i(U_i) \times \mathbb{C}$ modulo the identifications given by the transition functions defined as follows.

Now go to U_j and suppose the point is in the intersection. The same vector must be expressed as

$$\mu\left(\frac{z_0}{z_j}, \dots, \frac{z_{j-1}}{z_j}, 1, \frac{z_{j+1}}{z_j}, \dots, \frac{z_n}{z_j}\right).$$

To pass from one expression of p to another we multiply by

$$\frac{z_i}{z_j}$$

and thus we want our transition functions

$$g_{ij} : U_i \cap U_j \to \mathrm{GL}(1, \mathbb{C})$$

to satisfy

$$\mu = \lambda \frac{z_j}{z_i}$$

Obviously we take

$$g_{ij}(p) = \frac{z_j}{z_i}.$$

It's clear these satisfy the cocycle condition.

Definition 6.7.

$$\mathcal{O}(1) := \mathcal{O}(-1)^\vee, \qquad \mathcal{O}(d) := \bigotimes_d \mathcal{O}(1).$$

To show that the sections of $\mathcal{O}(d)$ are degree-d homogeneous polynomials we first show that the fibers are degree-d homogeneous functions (not necessarily polynomials) and then pass to the global situation.

Lemma 6.8. *The fiber $\mathcal{O}(d)_p$ is in bijection with degree-d homogeneous functions on p.*

Proof. So far we can only say that the fibers of $\mathcal{O}(-1) \otimes \mathcal{O}(1)$ are the fibes of $\mathcal{O}$. This means that for a fixed $f \in \mathcal{O}(1)_p$ we have a linear functional $z \mapsto f \otimes z$ on $\mathcal{O}(-1)_p$. Any linear functional on $\mathcal{O}(-1)_p$ can be realised this way.

To generalize this for degree d we do as follows. First note that

$$\mathrm{Hom}(\mathcal{O}(-1)_p, \mathbb{C}) \otimes \mathrm{Hom}(\mathcal{O}(-1)_p, \mathbb{C}) \cong \mathrm{Hom}(\mathcal{O}(-1)_p \otimes \mathcal{O}(-1)_p, \mathbb{C}).$$

(This is not true for general modules, but in this case it is using a basis.)

By the universal property of tensor product (see Commutative Algebra Theorem [**?**]) we know that $\mathrm{Hom}(\mathcal{O}(-1)_p \otimes \dots \otimes \mathcal{O}(-1)_p, \mathbb{C})$ is the set of multilinear maps $\mathrm{Lin}(\mathcal{O}(-1)_p \times \dots \times \mathcal{O}(-1)_p, \mathbb{C})$. Any such map L gives a degree-d function (a polynomial) P by $P(x) := L(x, \dots, x)$. In fact, since the fiber $\mathcal{O}(-1)_p$ is one-dimensional, we can reconstruct L from P by defining the multilinear map on a basis (which is composed by a single vector x) by $L(x, \dots, x) := P(x)$. This independs on the choice of basis since if $x' = \lambda x$ and we define $L'(x', \dots, x') := P(x')$, We see that $\lambda^d L'(x, \dots, x) = L'(x', \dots, x') == P(x') = P(\lambda x) = \lambda^d P(x) = \lambda^d L(x, \dots, x)$. $\square$

To pass from the local situation in Lemma **??** to a section we do as follows.

Exercise 6.9. Show that any non-trivial homogeneous degree-d polynomial $0 \neq f \in \mathbb{C}[z_0, \dots, z_n]_d$ can be considered as a non-trivial section of $\mathcal{O}(d)$ on $\mathbb{P}^n$.

Proof. Let P be a homogeneous degree-d polynomial. We define a section fiberwise and then show it is holomorphic. For $p \in \mathbb{P}^n$ we restrict P to the 1-dimensional vector space that p is in $\mathbb{C}^{n+1}$. To show this is holomorphic we can evaluate our section on a local frame for $\mathcal{O}(d)$, which gives a holomorphic function since P is a polynoimal (see [Lee24, Proposition 3.35]. □

Theorem 6.10. *Sections of $\mathcal{O}(d)$ are in correspondence with homogeneous degree-d polynomials.*

Proof. It only remains to prove the direct implication. Let $s \in H^0(\mathcal{O}(d))$. Then $s(p) \in \mathcal{O}(d)_p$ is a homogeneous degree-d function f_p by Lemma 6.8 defined on the vector space p.

Now define a holomorphic homogeneous degree-d function f on $\mathbb{C}^{n+1} \setminus \{0\}$ as follows. For any $z \in \mathbb{C}^{n+1} \setminus \{0\}$ we can take p to be the line passing through z and assign the number $f_p(z)$ (since $z \in p$).

Finally we show that f must be a polynomial by considerations on its Taylor series after extending to $\mathbb{C}^{n+1}$ via Hartog's theorem (see [Lee24, Theorem 3.36]). □

Exercise 6.11. Let L be a holomorphic line bundle on a compact complex manifold X. Show that L is trivial if and only if L and its dual $L^\vee$ admit non-trivial global sections.

Proof. Direct implication is easy: trivial bundle has 1-dimensional space of sections. For the converse choose nonvtrivial sections s of L and $s^\vee$ of $L^\vee$. Then $s \otimes s^\vee$ which is a section of the trivial bundle. If such a section has a zero, then it must be zero throughout X. □

Now let's discuss Serre's duality. It is a consequence of Hodge theorem, which says that Hodge star operator descends to cohomology. In fact, the general result is

Theorem 6.12 (Hodge)**.**

$$* : H^{p,q}(X) \longrightarrow H^{n-p,n-q}(X)$$

is an isomorphism.

However, we shall not use this isomorphic directly for Serre duality, see [Huy05, Remark 4.1.17]. What we do instead is show that the pairing of forms given by integration is nondegenerate. The upshot is that the proof of nondegeneracy is given by the Hodge star operator in cohomology.

Proposition 6.13 (Serre duality)**.** *Let X be a compact complex manifold. For any holomorphic vector bundle E on X, the natural pairing*

$$H^{p,q}(X, E) \times H^{n-p,n-q}(X, E) \to \mathbb{C}$$

is nondegenerate.

Proof. We must show that for any $\alpha \in H^{p,q}(X, E)$ there exists $\beta \in H^{n-p,n-q}(X, E)$ such that

$$\int_X \alpha \wedge \beta \neq 0.$$

Choose

$$\beta = \overline{*}\beta$$

where $\overline{*}$ is the conjugate of the Hodge star operator in cohomology. □

Of course, this proposition lacks context in these notes; we should properly extend Hodge theory to sections of arbitrary sections of vector bundles; but it looks as if the theory for sections of the cotangent algebra generalizes without the need of introducing any new ideas.

We put $p = 0$ to obtain

$$H^{0,q}(X) = H^q(X, \Omega^0) = H^q(X, \mathcal{O})$$

and

$$H^{n,n-q}(X) = H^{n-q}(X, \Omega^n).$$

Definition 6.14. Let (M, I, g) be a hermitian manifold. A linear connection ∇ on TM is *hermitian* if $\nabla g = 0$ and $\nabla J = 0$.

7. Line bundles and divisors

A simple way of associating a line bundle to an irreducible hypersurface $Y \subset X$ is by noting that by Weierstrass Division Theorem **??** the ideal sheaf of Y is locally generated, i.e. there is an open cover U_i of X such that $\mathcal{I}(U_i) = (f_i)$. This means that at intersections we have $f_j \in (f_i)$ and $f_i \in (f_j)$, which says that f_i and f_j are associates, i.e. there is a unit $u_{ij} \in \mathcal{O}_X^*(U)$ (the units in the ring of holomorphic functions are nowhere vanishing functions) such that $f_i = u_{ij} f_j$. Thus $u_{ij} : U_i \cap U_j \to \mathrm{GL}(1, \mathbb{C})$ define a cocycle in $H^1(X, \mathcal{O}_X^*) = \mathrm{Pic}(X)$.

Definition 7.1. A *Weil divisor* is a formal sum

$$D = \sum a_i Y_i$$

where $a_i \in \mathbb{Z}$ and Y_i are irreducible hypersurfaces.

An irreducible hypersurface Y clearly defines a Weil divisor, and so does a reducible reduced hypersurface with irreducible components Y_i putting $a_i = 1$. To allow for coefficients > 1 we may admit nonreduced hypersurfaces, so that we have larger multiplicites.

This can also be presented as follows. Given a holomorphic function g, we define *order of vanishing of g at $P \in Y$* to be the largest number $\mathrm{ord}_{P,Y} g$ such that

$$g = f_i^{\mathrm{ord}_{P,Y} g} h, \qquad h \in \mathcal{O}_{X,P}^*,$$

where f_i is an irreducible function vanishing along Y near P, i.e. a generator of the local ring of Y. Further, by coherence/preservation relative primes in nearby stalks (see [**?**, p. 696]) the order of g is constant for any $P \in Y$.

Thus for any holomorphic function g we can define the divisor

$$\mathrm{div}(g) := \sum_Y \mathrm{ord}_Y(g) Y$$

where the sum ranges in all irreducible hypersurfaces of X.

Similarly, for a meromorphic function $g = g_1/g_2$ we define the order of vanishing of g at Y to be

$$\mathrm{ord}_Y(g) = \mathrm{ord}_Y(g_1) - \mathrm{ord}_Y(g_2).$$

The associated divisor to a meromorphic function is defined identically as that of a holomorphic function, and we observe that there might appear negative coefficients. Divisors obtained in this way from meromorphic functions are called *principal divisors.*

This shows how to assign a divisor from any meromorphic function, i.e. it's a map

$$H^0(X, \mathcal{M}^*) \to \mathrm{Div}(X).$$

Note that if g is in fact a nowhere vanishing meromorphic or holomorphic function, the corresponding divisor is trivial, i.e. it is in the kernel of our map. In fact,

$$H^0(X, \mathcal{M}^*/\mathcal{O}^*) \simeq \mathrm{Div}(X).$$

Indeed, given a divisor we can assign a section of $\mathcal{M}^*/\mathcal{O}_X^*$ as follows. If the divisor is an irreducible hypersurface locally given by $V(f_i)$ we assign the meromorphic function given locally as f_i. If the divisor has several irreducible connected components we consider the product of the irreducible generators at every point.

It remains to connect $\mathrm{Div}(X)$ and $\mathrm{Pic}(X)$. The idea is that a section f of $\mathcal{M}^*/\mathcal{O}_X^*$ defines a cocycle in $H^1(\mathcal{O}_X^*)$. Indeed, f is given, by definition of quotient sheaf (?), by the data of an open cover U_i and meromorphic functions

$$f_i = \frac{g_i}{h_i} \in \mathcal{M}^*$$

which coincide at the intersections $U_i \cap U_j$, that is,

$$f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}.$$

But since we are in the quotient ring, equality means that

$$f_i f_j^{-1} \in \mathcal{O}_X^*.$$

This defines a cocycle, and thus a map

$$\mathrm{Div}(X) \to \mathrm{Pic}(X).$$

This map can also be though of as induced from

$$0 \longrightarrow \mathcal{O}_X^* \longrightarrow \mathcal{M}^* \longrightarrow \mathcal{M}^*/\mathcal{O}_X^* \longrightarrow 0$$

via the cohomology long exact sequence giving

$$\cdots \longrightarrow H^0(\mathcal{M}_X^*) \longrightarrow H^0(\mathcal{M}_X^*/\mathcal{O}_X^*) \longrightarrow H^1(\mathcal{O}_X^*) \longrightarrow \cdots$$

Indeed, the kernel of the map $\mathrm{Div}(X) \to \mathrm{Pic}(X)$ is given by the principal divisors, i.e. those coming from meromorphic sections.

In general there is no way to ensure this map is an isomorphism.

Now we discuss how to obtain submanifolds from sections. We will see that $\mathrm{Div}(X)$ is isomorphic to the group of line bundles equipped with a meromorphic section.

Bear in mind the following observation for a rank-r vector bundle $E \to X$. Suppose (φ_i, U_i) are local trivializations of E, i.e. $\varphi_i : \pi^{-1}(U_i) \xrightarrow{\simeq} U_i \times \mathbb{C}^k$. Take a local section $s : U_i \to \pi^{-1}(U_i)$. Take a local basis e_i of $\mathbb{C}^k$. Then

$$s_i(x) = f_i^k(x)\varphi_i^{-1}(x, e_k)$$

for some functions $f_i^k : U_i \to \mathbb{C}$.

At an intersection $U_i \cap U_j$, we have

$$s_i(x) = f_i^k(x)\varphi_i^{-1}(x, e_k) = s_j(x) = f_j^k(x)\varphi_j(x, e_k).$$

Applying φ_i we see that

$$f_i^k e_k = f_j^k g_{ij}(e_k).$$

In the case of a line bundle this just says that sections are given by local functions $f_i : U_i \to \mathbb{C}$ satisfying
$$f_i = g_{ij} f_j$$
where g_{ij} is the cocycle defining the line bundle.

Let L be a line bundle. A *meromorphic section* of L is a section of $\mathcal{O}(L) \otimes_{\mathcal{O}} \mathcal{M}$, which is given by a collection of meromorphic functions $s_i \in \mathcal{M}(U_i)$ such that
$$s_i = g_{ij} s_j.$$
That is, sections of the line bundle viewed as a module over $\mathcal{O}$ with coefficients that can be meromorphic functions.

The orders of the coefficient meromorphic functions is well defined along hypersurfaces of X, and thus we can define the *divisor div*(s) *associated to the section* s *of* L by
$$\operatorname{div}(s) := \sum_Y \operatorname{ord}_Y(s) Y.$$

Conversely, given any divisor D we can assign a line bundle L as before, namely if the divisor is given by the meromorphic functions f_i we put $g_{ij} = f_i/f_j$. But now we observe the existence of a distinguished section s such that $\operatorname{div}(s) = L$. Indeed, we note that
$$f_i = f_i f_j^{-1} f_j = g_{ij} f_j,$$
which is a section of L, and in fact (check...) $\operatorname{div}(s) = L$.

We conclude that L is the line bundle associated to a divisor if and only if it has a meromorphic section not identically zero. It is the line bundle associated to an effective divisor if and only if it has a holomorphic section not identically zero — whose zero set is the divisor.

Upshot. We have more than the submanifold associated to a section: we have the divisor associated to a meromorphic section, which gains to becoming less geometrically intuitive the virtue of being an element of a group.

Now we turn to line systems. Take a divisor
$$D = \sum a_i V_i.$$
The divisor
$$D + (f)$$
is linearly equivalent to D — their difference is principal. If
$$D + (f) \geq 0,$$
then f must be holomorphic away from V_i, and on V_i it can have poles with the condition
$$\operatorname{ord}_{V_i}(f) \geq -a_i.$$
Denote $\mathcal{L}(D)$ the space of such functions.

Note that if s_0 is a section such that $(s_0) = D$, then multiplication by s_0 gives an identification
$$\mathcal{L}(D) \xrightarrow{\otimes s_0} H^0(X, L).$$

Definition 7.2. We denote by
$$|D| \subset \operatorname{Div}(X)$$
the set of all effective divisors linearly equivalent to D.

$\mathcal{L}(D) = H^0(X, L)$ is almost the same thing as $|D|$. At least in the compact case, two functions in $f, f' \in \mathcal{L}(D)$ give the same divisor when $f = \lambda f'$ for some constant function $\lambda \in \mathbb{C}$.

Thus

$$|D| = \mathbb{P}H^0(X, L).$$

This is called a *complete linear system*. More generally, a *linear system* is a linear subspace of $\mathbb{P}H^0(X, L)$. The *base locus* of $|D|$ is the set of points where all the global sections generating $H^0(X, L)$ vanish. We will see in Section 12 the relationship between line systems and embeddings to projective space.

Theorem 7.3 (Bertini)**.** *The generic divisor of a linear system is smooth away from the base locus.*

Now let's introduce the first adjunction formula, which says that when a divisor is a smooth hypersurface, it's associated line bundle is its holomorphic normal bundle.

Theorem 7.4 (Adjunction formula I)**.** *If X is a smooth hypersurface of a smooth complex manifold Y, then*

$$\mathcal{O}_Y(X)|_X \simeq \mathcal{N}_{X/Y}, \tag{7.4.1}$$

where $\mathcal{N}_{X/Y} := T'_Y/T'_X$.

Proof. If X is locally defined by the vanishing of f_i on an open cover U_i of Y, then $\mathcal{O}_Y(X)$ is given by the cocycle $f_{ij} := f_i f_j$.

To prove Equation 7.4.1 it's enough to show that $\mathcal{O}_Y(X)|_X \otimes \mathcal{N}^\vee_{X/Y}$ is trivial, which can be done by constructing a nonvanishing section. Such a section is given by

$$df_i = d(f_{ij} f_j) = df_{ij} \cdot f_j + f_{ij} df_j,$$

since the first term vanishes along X (because f_j does) and the remaining terms define precisely a section of $T'^\vee_Y/T'^\vee_X$, the conormal bundle — indeed, sections of such a bundle are holomorphic differencials vanishing along X. □

Now let's find the canonical bundle of $\mathbb{P}^n$. By the above discussion, we need the inverse map of

$$\mathrm{Div}(\mathbb{P}^n) \to \{L \in \mathrm{Pic}(X) : L \text{ has a nonzero meromorphic section}\},$$

that is, we need to find a nonzero meromorphic section s of the canonical bundle $\Omega^{n,0} = K$. Then the divisor will be given by

$$\sum_Y \mathrm{ord}_Y s[Y].$$

A meromorphic section is given by local sections s_i so that

$$s_i = g_{ij} s_j$$

where g_{ij} is the cocycle of K.

Consider in the canonical chart $z_0 \neq 0$ of $\mathbb{P}^n$ the local meromorphic section of K

$$s_0 = \frac{dw_1}{w_1} \wedge \frac{dw_2}{w_2} \wedge \ldots \wedge \frac{dw_n}{w_n}$$

where $w_i = z_i/z_0$ are the coordinate functions in $z_0 \neq 0$.

In any chart $z_j \neq 0$ we obtain the local expression

$$(-1)^j \frac{dw'_0}{w'_0} \wedge \ldots \wedge \widehat{\frac{dw'_j}{w'_j}} \wedge \ldots \wedge \frac{dw'_n}{w'_n}$$

according to the canonical change of coordinates, which says

$$w_i = \frac{w'_i}{w'0}, i \neq j, \qquad w_j = \frac{1}{w'_0}.$$

The point is that we constructed a meromorphic section which has a single pole along each hyperplane $z_i = 0$, which means

$$K = -(n+1)H.$$

The next adjunction formula is the key to computing the canonical bundle of submanifolds. It is a consequence of this linear algebra fact:

Lemma 7.5. *If*

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0$$

is a short exact sequence of vector bundles (of possibly different ranks), then

$$\det(E) \cong \det(E') \otimes \det(E'').$$

Proof. First notice that each of the three determinant bundles gives fiberwise the top-exterior power of the corresponding vector spaces, thus varying the rank of the exterior power taken. Then define a map

$$x_1 \wedge \ldots \wedge x_{\mathrm{rk}E'} \otimes x_{\mathrm{rk}E'+1} \wedge \ldots \wedge x_{\mathrm{rk}E'+\mathrm{rk}E''}$$
$$\mapsto x_1 \wedge \ldots \wedge x_{\mathrm{rk}E'} \wedge x'_{\mathrm{rk}E'+1} \wedge \ldots \wedge x'_{\mathrm{rk}E'+\mathrm{rk}E''}$$

where $x'_{\mathrm{rk}E'+i}$ maps to $x_{\mathrm{rk}E'+i}$ under the surjection given by the exact sequence.

It's clear that such a map is injective: if any monomial in the image is zero, then one of the factors must be zero, and then the element we began with also vanishes. Being two vector spaces of dimension 1, the map must also be surjective. Check this map is well-defined! This gives an isomorphism in every fiber.

In fact the isomorphism above acts like 1. To show that this glues to a bundle isomorphism we have to check some cocycle condition as in Proposition 6.3. In this case the transformation functions A_i are just 1 by definition of the morphism. The cocycle condition follows from the fact that any short exact sequence splits, so on U_{ij} we have the isomorphism $E(U_{ij}) \simeq E'(U_{ij}) \oplus E''(U_{ij})$. Recall that the cocycle of the direct sum of bundles is given by $\begin{pmatrix} h'_{ij} & 0 \\ 0 & h''_{ij} \end{pmatrix}$ so that when taking determinants we obtain that $g_{ij} = h'_{ij}h''_{ij} := h_{ij}$ (since taking determinant bundle is taking determinant of the transition functions). Thus the transition functions of the two bundles in question actually coincide, and the cocycle condition in Proposition 6.3 is the trivial equality $g_{ij} = h_{ij}$. $\square$

Here's a cleaner version of this proof:

Proof. Choose an open cover $\{U_i\}$ on which the sequence splits, so that

$$E|_{U_i} \simeq E'|_{U_i} \oplus E''|_{U_i}.$$

Pick local frames $(e'_{i,1}, \ldots, e'_{i,r})$ for $E'|_{U_i}$ and $(e''_{i,1}, \ldots, e''_{i,s})$ for $E''|_{U_i}$, and use the splitting to obtain a frame $(e'_{i,\bullet}, e''_{i,\bullet})$ of $E|_{U_i}$.

On overlaps U_{ij} the transition matrix of E in these adapted frames has block diagonal form

$$g_{ij}^E = \begin{pmatrix} g'_{ij} & 0 \\ 0 & g''_{ij} \end{pmatrix},$$

hence

$$\det(g_{ij}^E) = \det(g'_{ij})\,\det(g''_{ij}).$$

Thus the transition functions of $\det(E)$ coincide with those of $\det(E') \otimes \det(E'')$, giving an isomorphism of line bundles. In these frames it is given by

$$(e'_{i,1} \wedge \cdots \wedge e'_{i,r}) \otimes (e''_{i,1} \wedge \cdots \wedge e''_{i,s}) \longmapsto e'_{i,1} \wedge \cdots \wedge e'_{i,r} \wedge e''_{i,1} \wedge \cdots \wedge e''_{i,s}.$$

If a different local splitting is chosen, the adapted frames differ by a matrix of the form

$$\begin{pmatrix} I & B \\ 0 & I \end{pmatrix},$$

whose determinant is 1. Therefore the induced map on determinant lines is unchanged, showing that the above isomorphism is independent of the splitting and hence canonical. □

Theorem 7.6 (Adjunction formula II)**.** *Let X be a complex submanifold of a complex manifold Y. Then*

$$(K_Y \otimes \mathcal{O}_Y(X))|_X \cong K_X. \tag{7.6.1}$$

Proof. From the holomorphic tangent bundle exact sequence we get

$$0 \longrightarrow T'_X \longrightarrow T'_Y|_X \longrightarrow N_{X/Y} := T'_Y/T'_X \longrightarrow 0$$

Taking determinant of this exact sequence we obtain the result. □

Now we turn to Chern classes of line bundles.

Slogan: the first Chern class of a line bundle is the Poincaré dual of the divisor's fundamental class and the cohomology class of the curvature form of any connection of L. It is an integral $(1,1)$-curvature form.

The *curvature form* is the differential 2-form that represents the curvature operator with respect to a given connection on the line bundle. This can also be defined for affine vector bundles of any rank.

Proposition 7.7. (1) *For any line bundle L with curvature form Θ,*

$$c_1(L) = \left[\frac{\sqrt{-1}}{2\pi}\Theta\right] \in H^2_{dR}(X).$$

(2) *If $L = [D]$ for some $D \in Div(M)$, $c_1(L)$ is the Poincaré dual of the fundamental class of D.*

Finally let's just put some more exercises here.

Exercise 7.8 (Curves on surfaces and the Hodge index theorem)**.** Let S be a smooth projective surface and let $\phi : S \to \mathbb{P}^n$ be a holomorphic map satisfying the following properties:

- There exists a curve $C \subset S$ such that $\phi(C)$ is a finite set of points. (Here C is a not necessarily connected finite union of irreducible hypersurfaces C_i which are not necessarily smooth.)

- There exists an open set of S where ϕ is an embedding.

(1) Show that the class $h = \phi^* c_1(\mathcal{O}_{\mathbb{P}^n}(1))$ satisfies $\langle h, h\rangle > 0$ where $\langle\cdot,\cdot\rangle$ is the intersection form on $H^2(S,\mathbb{Z})$.
(2) Show that $\langle [C_i], h\rangle = 0$ for any i. Deduce from the Hodge index theorem that the intersection form $\langle\cdot,\cdot\rangle$ is negative definite on the subgroup of $H^2(S,\mathbb{Z})$ generated by the $[C_i]$'s.
(3) Show that $\langle C_i, C_j\rangle \geq 0$ for $i \neq j$. Deduce then from the above that the $[C_i]$'s are independent in $H^2(S,\mathbb{Z})$.

Proof. (1) Here's an outline of how to prove this result. First notice that h is the class of a line bundle since $\varphi^* c_1 = c_1 \varphi^*$. Thus $\langle h, h\rangle$ is the intersection number of two divisors in the class h.

To compute their intersection number we use that φ is an embedding on an open set of S, say $U \simeq \varphi(U) \subset \mathbb{P}^n$. To compute the intersection number we take two generic hyperplanes H_1 and H_2 in $\mathbb{P}^n$ and intersect them with $\phi(U)$. The dimension of the resulting submanifold is zero. Thus the intersection is a finite number of points.

(2) Since C_i is mapped to a set of points in $\mathbb{P}^n$, we compute the intersection number as above, which gives zero for generic choice of H_1, H_2. Hodge index theorem tells us that the signature of $\langle\cdot,\cdot\rangle$ is $(1, \rho - 1)$ on $H^2(S,\mathbb{Z})$. Since h is positive and C_i is orthogonal to h, it's necessary that C_i are negative vectors.

(3) This is by definition/construction of the intersection form on a surface: for two irreducible divisors of the surface, the intersection number is the number of intersection points of the divisors counted with multiplicity. Such a number cannot be negative. The fact that the C_i's are independent follows from a linear-algebra argument.

□

8. Kodaira Vanishing theorem

In [**?**, p. 150] we see that the curvature form of the tautological bundle is given by $\partial\bar{\partial}\log|z|^2$ where z is a nonzero section. Not giving further details for now, this implies that the Chern class of the hyperplane bundle is the Fubini-Study form in $\mathbb{P}^n$. We already discussed that this form is positive in the sense of Definition 3.8.

Definition 8.1. Let M be a compact Kähler manifold. A line bundle $L \to M$ is *positive* if there exists a metric on L with curvature form Θ such that $(\sqrt{-1}/2\pi)\Theta$ is a positive $(1,1)$-form.

Theorem 8.2 (Kodaira-Nakano Vanishing Theorem)**.** *[Let M be a compact Kähler manifold of complex dimension n.] If $L \to M$ is a positive line bundle, then*

$$H^q(M, \Omega^p(L)) = 0, \qquad p + q > n = \dim_{\mathbb{C}} M.$$

Here's the same result as it appears in [**?**, **?**, sea] where it is not proved.

Theorem 8.3 (Kodaira Vanishing)**.** *Suppose k is a field of characteristic 0, and X is a smooth projective k-variety. Then for any ample invertible sheaf L, $H^i(X, K_X \otimes L) = 0$ for $i > 0$.*

For completeness we put another vanishing theorem concerning higher-rank vector bundles (probably due to Serre though Griffith-Harris don't say so). The idea is that twisting an ample line bundle with itself enough times and tensoring with any vector bundle E will kill positive cohomology.

Theorem 8.4. *Let M be a compact, complex manifold and $L \to M$ a positive line bundle. Then for any holomorphic vector bundle E, there exists μ such that*

$$H^q(M, L^\mu \otimes E) = 0 \qquad \text{for } q > 0, \mu \geq \mu_0.$$

9. Lefschetz hyperplane sections theorem

Lefschetz hyperplane theorem holds for any n-dimensional compact complex manifold M and $V \subset M$ a smooth hypersurface with $L = [V]$ positive. An example of this is when M is projective and V is the pullback of the hyperplane bundle (which, as explained in Section 8, is positive since its curvature form is Fubini-Study), that is, $V = M \cap H$ for a generic hyperplane $H \subset \mathbb{P}^N$.

Theorem 9.1 (Lefschetz hyperplane sections)**.** *The map*

$$H^q(M, \mathbb{Q}) \to H^q(V, \mathbb{Q})$$

induced by the inclusion $V \hookrightarrow M$ is an isomorphism for $q \leq n-2$ and injective for $q = n-1$.

Proof. Looks accessible! "Purely topological". □

10. Complex varieties

For a definition of sheaf see Algebraic Geometry Definition 12.1.

Lemma 10.1. *Let X be a complex manifold of complex dimension n.*
The functor

$$\begin{aligned} \mathcal{O}_X : Open_X^{op} &\longrightarrow Set \\ U &\longmapsto \mathcal{O}_X(U) := \{f : U \to \mathbb{C}^n : f \text{ is holomorphic}\} \\ i : V \hookrightarrow V &\longmapsto \mathcal{O}_X(i) : \mathcal{O}_X(U) \to \mathcal{O}_X(V) \end{aligned}$$

where $\mathcal{O}_X(i)$ is given by restriction, is a sheaf.

Proof. $\mathcal{O}_X$ is a presheaf, i.e. a functor, by definition: restriction maps the identity to the identity and preserves compositions.

To check the condition of being a sheaf let U_i be an open cover of some open set U of X and $f_i : U_i \to X$ sections of $\mathcal{O}_X(U_i)$, that is, holomorphic functions, which coincide pairwise in intersections. Then it is obvious that the function $f : U \to \mathbb{C}$, $x \mapsto f_i(x)$ for any $U_i \ni x$ is well defined. It is also holomorphic since holomorphicity is a condition defined on open sets. □

A meromorphic function is a function that can be written locally as a quotient of two holomorphic functions:

Definition 10.2. A *meromorphic function* on a complex manifold X is a map

$$f : X \to \coprod_{x \in X} \operatorname{Frac}(\mathcal{O}_{X,x})$$

which associates to any $x \in X$ an element $f_x \in \operatorname{Frac}(\mathcal{O}_{X,x})$ such that for any $x_0 \in X$ there exists $U \ni x$ open and $f, g \in \mathcal{O}(U)$ such that $f_x = g/h$ for all $x \in U$. The *sheaf of meromorphic functions* is $\mathcal{K}_X$.

Now we pass to the theory of analytic subvarieties.

Definition 10.3. An *analytic set* or *subvariety* Y is a subset of a complex manifold X given locally as the zeros of a finite collection of holomorphic functions.

Lemma 10.4. *Let $Y \subset X$ be an analytic subvariety of a complex manifold X equipped with the subspace topology. Then*

$$\begin{aligned} \mathcal{O}_Y : Open_Y^{op} &\longrightarrow Set \\ U \cap Y &\longmapsto \{f|_{U\cap Y} : f \in \mathcal{O}_X(U)\} \end{aligned}$$

is a sheaf.

Proof. This is a presheaf for the same reasons as the sheaf $\mathcal{O}_X$ is a presheaf. The fact that it is a sheaf follows from the fact that $\mathcal{O}_X$ is a sheaf and by definition of subspace topology; namely for any family of functions defined over an open cover $\{U_i \cap Y\}$ of some open set $U \cap Y$ we can reconstruct a function on U that restricted to $U \cap Y$ is as desired. □

Recall that any map of sheaves of groups has well-defined kernel, see Lemma **??**.

Definition 10.5. Let $Y \subset X$ be an analytic subvariety of a complex manifold X. The *ideal sheaf* of Y, denoted $\mathcal{I}_Y$, is the kernel of the restriction map $\mathcal{O}_X \to \mathcal{O}_Y$.

This just says that $\mathcal{I}_X(U)$ is the set of functions $f \in \mathcal{O}_X$ that vanish under the restriction map, i.e. that vanish at points of Y.

Since the restriction map is surjective, we have the following short exact sequence:

$$0 \longrightarrow \mathcal{I}_Y \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_Y \longrightarrow 0$$

Notice that for any open set $U \subset X$, $\mathcal{I}_Y(U)$ is an ideal of $\mathcal{O}_X(U)$. (Indeed, if a function vanishes at Y then its product with any other function will also vanish.)

Definition 10.6. An analytic variety $V \subset U \subset \mathbb{C}^n$ is *irreducible* if V cannot be written as the union of two distinct analytic varieties $V_1, V_2 \subset U$, both distinct to V.

We will show that for the case of an irreducible hypersurface the ideal sheaf is locally generated by a single element. This will allow us to construct a line bundle from the defining functions of the hypersurface, which is in general not possible for an arbitrary smooth hypersurface, — although it suffices to ask for orientability in the smooth case.

We need two ingredients: (1) the fact that the stalks $\mathcal{O}_p$ of the sheaf of holomorphic functions are UFDs, and (2) the following lemma.

Lemma 10.7. *The ideal (correction: this should be the stalk) of an irreducible analytic hypersurface is a height-1 prime ideal.*

Proof. We already know that irreducible varieties correspond with prime ideals.

To show the ideal I defining the variety Y has height 1 suppose that $0 \subset \mathfrak{p} \subseteq I$ is prime. Then $V(\mathfrak{p})$ is an analytic variety that contains Y. At regular points of both $V(\mathfrak{p})$ and Y, both are smooth manifolds, but since Y is of codimension 1 and $V(\mathfrak{p})$ does not equal all of Y, we conclude that they coincide. Thus, in a neighbourhood of a regular point we have $\mathfrak{p} = I(V(\mathfrak{p})) = I$ by the Nullstellensatz. □

Lemma 10.8. *The stalks of the ideal sheaf of an irreducible hypersurface are principal rings.*

Proof. Since $\mathcal{O}_p$ is a UFD, we pick any element $f \in \mathcal{I}_p$, and factor it as a product of irreducibles. Since $\mathcal{I}$ is prime, some of the irreducible factors, say f_i, is in $\mathcal{I}_p$. The ideal generated by f_i is prime since $\mathcal{O}_p$ js a UFD (see 034T). But by Lemma 10.7, $\mathcal{I}_p$ is a height-1 prime ideal, so the prime ideal (f_i) must equal all of $\mathcal{I}_p$. $\square$

Lemma 10.9. *The ideal sheaf $\mathcal{I}$ of the irreducible hypersurface $Y \subset X$ corresponds to a unique line bundle.*

Proof. Let $p \in U$ and consider for any stalk $\mathcal{I}_p$ its generator $[f_p]$, which has a representative (f_p, U_p). These form an open cover of X, coincide up to a unit in the intersections as follows. These units shall give us a cocycle.

For any $U_i \cap U_j \neq 0$ we have [This is slightly imprecise. We may extend $f_j|_{U_i \cap U_j}$ to U_i and $f_i|_{U_i \cap U_j}$ to U_j, and then restrict back to $U_i \cap U_j$. This works because we can extend holomorphic functions by their local power series expansion!], $f_j \in (f_i)$, so there exist functions g_{ij} such that

$$f_i = g_{ij} f_j.$$

These satisfy the cocycle condition since

$$f_i f_j f_k = g_{ij} f_j g_{jk} f_k g_{ki} f_i.$$

By Proposition 6.2 we have a line bundle. $\square$

Finally we have concluded that the ideal sheaf of a hypersurface is a line bundle. Now we can take its dual to define:

Definition 10.10. Let $D := S$ be an effective Cartier divisor, that is, an analytic hypersurface of a complex manifold. The line bundle $\mathcal{O}(D)$ is the dual line bundle of the ideal sheaf of D.

11. Chow's theorem

Theorem 11.1 (Chow)**.** *Any analytic subvariety of $\mathbb{C}P^n$ is algebraic.*

Proof. First we prove it for hypersurfaces. A hypersurface in $\mathbb{P}^n$ defines a line bundle along with a section vanishing on the hypersurface. Such a line bundle must be $\mathcal{O}(d)$ for some d by Exercise 4.13. Sections of $\mathcal{O}(d)$ are homogeneous degree-d polynomials by Theorem 6.10.

For higher codimension the challenge is find adequate planes in $\mathbb{P}^n$, so that we can project V, $\dim V = k$ to $\mathbb{P}^{k+1}$ and apply the hypersurface version. Pick any point $p \in \mathbb{P}^n$ not in V. We can choose a $(n-k-1)$-plane not intersecting V and containing p. Inside such $\mathbb{P}^{n-k-1}$ we can pick a $\mathbb{P}^{n-k-2}$ not touching p. Now pick a complementary $\mathbb{P}^{k+1}$ in $\mathbb{P}^n$, i.e. $\mathbb{P}^n = \mathbb{P}^{k+1} \oplus \mathbb{P}^{n-k-2}$.

Now project V to $\mathbb{P}^{k+1}$ and apply Chow for hypersurfaces. We obtain a polynomial F defining the projection of V. Extend F to $\tilde{F}$ on all $\mathbb{P}^n$ by the rule $\tilde{F}(x_0, \ldots, x_n) = F(x_0, \ldots, x_{k+1})$. Note that since p is not in V, $\tilde{F}$ does not vanish in p. Thus V is the intersection of all such polynomials F, one for each p, which ensures that no points other than those of V are in the common zeroes. $\square$

12. Kodaira Embedding theorem

We introduced line systems in Section 7. Let L be a holomorphic bundle on a complex manifold X. A point $x \in X$ is a *base point* of L if $s(x) = 0$ for all $s \in H^0(X, L)$ and the *base locus* $\mathrm{Bs}(L)$ is the set of all base points of L.

Let $s_0, \dots, s_N \in H^0(X, L)$ are a basis for some subspace $E \subset H^0(X, L)$. Define a map

$$\begin{aligned} \varphi_L : X \setminus \mathrm{Bs}(L) &\longrightarrow \mathbb{P}^N \\ x &\longmapsto [s_0(x) : \ldots : s_N(x)] \end{aligned}$$

which is well-defined since we excluded the base locus, and holomorphic since we chose holomorphic sections of L for the coordinates. Further, the pullback of the hyperplane bundle of $\mathbb{P}^N$ is L, i.e.

$$\varphi_L^* \mathcal{O}_{\mathbb{P}^N}(1) \simeq L|_{X \setminus \mathrm{Bs}(L)}$$

since $s_i = 0$ is the equation of a hyperplane in $\mathbb{P}^N$, so, it is a section of H. Its pullback under φ gives the section s_i whose associated divisor (s_i) corresponds to the line bundle L — the line bundle associated to a divisor associated to a section of the line bundle is the line bundle itself.

The question is: when is φ an embedding?

Definition 12.1. A line bundle L on a complex manifold is called *ample* if for some $k > 0$ and some linear system in $H^0(X, L^k)$ the associated map φ is an embedding.

Theorem 12.2 (Kodaira Embedding)**.** *Suppose M is a compact complex manifold. A holomorphic line bundle $L \to M$ is ample if and only if it is positive. Thus M is projective if and only if it admits a positive holomorphic line bundle.*

Proof. We must show that φ separates points and has surjective differential everywhere… ☐

Compare this theorem with Nakai-Moishezon Criterion 16.1.

Exercise 12.3. Let L be an ample bundle on a K3 surface M. Prove that $L^{\otimes 2}$ is globally generated (that is, for each $x \in M$ there exists a section $h \in H^0(L^{\otimes 2})$ which does not vanish in x).

Proof. As an outline for exposition:

1. Show that L has sections (by ampleness+Kodaira Vanishing+Riemann-Roch for surfaces), pick a section C and note that if $L^{\otimes 2}$ had base points they would be in C.
2. Using Kodaira Vanishing again we show that every section of the line bundle restricted to C comes from a section of the line bundle on M. Thus it's enough to find a nonvanishing section on the restricted bundle.
3. Then we aim to apply Riemann-Roch on C. If we can show that $h^0((L^{\otimes 2})^\vee \otimes \omega_C)$ and $h^0((L^{\otimes 2}(-p))^\vee \otimes \omega_C)$ both vanish, we apply Riemann-Roch and see that $h^0(L^{\otimes 2}) > h^0(L^{\otimes 2}(-p)$, which proves the result since that says that not every section is a section vanishing at $p \in C$.
4. To compute the vanishings of the h^0 in the previous step we show they have negative degrees (genus formula + degree of canonical bundle of a curve).

First we need to show that L has sections. To see this I will shot that since L is ample, $L^2 > 0$. Indeed, if mL is very ample we apply by Riemann-Roch, Kodaira Vanishing and the fact that $\chi(\mathcal{O}_X) = 2$ on a K3 to obtain $h^0(mL) = 2 + \frac{1}{2}m^2L^2$ which must be positive since mL has sections by being very ample. Notice L^2 cannot be zero as $h^0(mL) = 2$ would imply we have an embedding of a K3 surface into $\mathbb{P}^1$.

Applying again Riemann-Roch and Kodaira vanishing we see that $h^0(L) > 0$. Let C be a section of L.

Notice that if $L^{\otimes 2}$ had any base points, they would have to be on C. Indeed, since C is the vanishing set of a section $s \in H^0(L)$, we must have $s \otimes s \in H^0(L^{\otimes 2})$ vanishing in any base point of $L^{\otimes 2}$.

Then we consider the ideal exact sequence for C and tensor by $L^{\otimes 2}$ (which preserves left-exactness because line bundles are locally free of rank 1, that is, we are tensoring by the underlying ring) to obtain

$$0 \longrightarrow L^{\otimes 2}(-C) \longrightarrow L^{\otimes 2} \longrightarrow L^{\otimes 2} \otimes \mathcal{O}_C \longrightarrow 0$$

Notice that the term on the left is actually L since the ideal sheaf $\mathcal{O}_M(-C) = \mathcal{I}_C$ is dual to L because C is defined as the vanishing set of a section of L. Passing to cohomology we obtain

$$H^0(L^{\otimes 2}) \longrightarrow H^0(L^{\otimes 2} \otimes \mathcal{O}_C) \longrightarrow H^1(L)$$

where the latter term vanishes by Kodaira Vanishing because L is ample. This says that every section of $L^{\otimes 2}|_C$ is the restriction to C of a section of $L^{\otimes 2}$. In turn this shows that it's enough to show that the bundle $L^{\otimes 2}$ has no base points along C.

By the genus formula for (possibly singular) curves on smooth surfaces and by the fact that M is smooth, we see that $2p_a(C) - 2 = (L.L)$ so that

$$4p_a(C) - 4 = (2L.L) = \deg_C(L^{\otimes 2})$$

Notice that since $L^2 > 0$ we exclude the cases that $p_a(C) = 0, 1$.

To conclude pick a point $p \in C$. Saying that p is not a base point of $L^{\otimes 2}|_C$ is the same as saying that not every section of $L^{\otimes 2}|_C$ vanishes at p, that is, that $h^0(L^{\otimes 2}|_C(-p)) < h^0(L^{\otimes 2}|_C)$.

Now I will prove that since the degree of $L^{\otimes 2}|_C$ and $L^{\otimes 2}|_C(p)$ is greater than $2p_a(C)$ we have that $h^0(\omega_C \otimes (L^{\otimes 2}|_C)^\vee) = 0 = h^0(\omega_C \otimes (L^{\otimes 2}|_C(-p))^\vee)$. For this we need to see these line bundles as Weil divisors by simply taking sections, whose vanishing sets are finite sets of points with multiplicities. Then their degree is the sum of the multiplicities. This definition makes degree additive with respect to tensor product (we may take local sections and sum degrees). By Riemann-Roch on the curve C, we know that $\deg \omega_C = 2p_a - 2$. Then

$$\deg_C(\omega_C \otimes L^{\otimes 2}) = \deg_C(\omega_C) + \deg_C(L^{\otimes 2}) = 2p_a - 2 - (4p_a(C) - 4) < 0$$

and a similar computation works for $L^{\otimes 2}(-p)$ as it has the same degree of $L^{\otimes 2}$ minus 1. This implies that neither of these bundles can have sections since any section would provide a linearly equivalent effective divisor — degree is preserved under linear equivalence, and divisors given by sections are effective. Therefore $h^0(\omega_C \otimes (L^{\otimes 2}|_C)^\vee) = 0 = h^0(\omega_C \otimes (L^{\otimes 2}|_C(-p))^\vee)$ as claimed.

Finally we apply Riemann-Roch to $L^{\otimes 2}$ and $L^{\otimes 2}(-p)$ to obtain that $h^0(L^{\otimes 2}) > h^0(L^{\otimes 2}(-p))$. $\square$

13. Riemann-Roch formulas

We want to know the dimension of $H^0(X, \mathcal{O}(D))$, where X is a curve and D is a divisor on X. To refresh out memory recall that if D is a smooth hypersurface (in this case this just means that it's a set of points without multiplicities) then that $\mathcal{O}(D)$ is given by the cocycle $g_{ij} = f_j/f_i$ where $D = \mathrm{div}(f_i)$ on U_i. The g_{ij} give a cocycle, which in turn is a line bundle on X. If e_i is a local basis then we have $e_i = g_{ij}e_j$ by definition of transition function. A section is given locally by h_ie_i and it has to glue, that is, $h_je_j = h_ie_i = e_if_j/f_ie_j$ so $h_j/f_j = h_i/f_i$. That's literally the data of a meromorphic function, namely some local meromorphic functions which coincide in intersections.

What's the vanishing of this meromorphic function along D? Now suppose D is an arbitrary Weil divisor, so the f_i have some multiplicities. If h_i has no zeroes, then the multiplicity of the new meromorphic function remains unaltered, but h_i can also vanish along D and then the vanishing becomes as positive as we want it. But it cannot have poles, the least positive it can be is the prescribed negativity of D.

This is why looking for $h^0(D)$ is just as much as looking for meromorphic functions h on X such that

$$\mathrm{div} h + D \geq 0 \qquad \text{(is effective).}$$

Theorem 13.1 (Riemann-Roch for curves)**.** *Let D be a divisor on a compact Riemann surface X, that is, D is a collection of points with multiplicities on X.*

$$h^0(D) - h^0(K - D) = d - g + 1 \tag{13.1.1}$$

Exercise 13.2. Prove that if X is a complex smooth curve with $g \geq 2$ then $h^0(T_X) = 0$.

Proof. First you notice that the dimension of the holomorphic 1-forms is the genus. This is just because the genus p_a is defined as $h^1(\mathcal{O}_X)$, which is just $h^0(K_X)$ by Serre duality. Then you say well if you have a holomorphic vector field, pair it with the nonzero holomorphic 1-form. This gives a nonzero function, but it must be constant because X is compact. Then it is actually nowhere vanishing. This says the vector field cannot vanish anywhere, which means the tangent bundle of the curve is trivial. Apparently elliptic curves, i.e. $g = 1$ are the □

Theorem 13.3 (Riemann-Roch for line bundles on surfaces)**.** *Let L be a line bundle on a complex surface X, and $K_X = \Omega^2 X$ the canonical bundle of X. Then*

$$\chi(L) = \chi(\mathcal{O}_X) + \frac{1}{2}(L, L - K_X) \tag{13.3.1}$$

Lemma 13.4 (Genus formula)**.** *Let C be any curve on a surface S. Then*

$$2p_a(C) - 2 = (C.C - K_S) \tag{13.4.1}$$

Proof. If C is nonsingular we may use the normal bundle short exact sequence. If C is singular we need to use the ideal sheaf short exact sequence, which means we think of C either as a closed embedded subscheme of S or as an analytic subvariety. Taking Euler class we obtain

$$\chi(\mathcal{O}(-S)) + \chi(\mathcal{O}_S) = \chi(\mathcal{O}_X)$$

Since $p_a(C) := 1 - \chi(\mathcal{O}_C)$, we obtain that $p_a(C) = 1 - \chi(\mathcal{O}_S) - \chi(\mathcal{O}(-C))$. To compute $\chi(\mathcal{O}(-C))$ we use Riemann-Roch formula **??** for curves on surfaces, which gives

$$\chi(\mathcal{O}(-C)) = \chi(\mathcal{O}_S) + \frac{(\mathcal{O}(-C).\mathcal{O}(-C) + \omega_S^*)}{2}$$

Taking duals on both entries of the intersection form and substituting in our previous expression for $p_a(C)$ we obtain the result. □

Exercise 13.5. Let S be a singular, irreducible complex curve on a K3 surface. Prove that $(S.S) \geq 0$.

Proof. This is just an application of Eq. 13.4.1. Since $K_X = \mathcal{O}_X$ it suffices to show that $p_a(S)$ is not zero. This follows fact that S is singular, since its arithmetic genus is defined as the genus of its normalization, which must be strictly positive since there exists at least one singularity. □

14. Hypercomplex manifolds

Exercise 14.1. Let M be a compact hypercomplex manifold of real dimension 4, equipped with a quaternionic Hermitian structure, and V the space of closed SU(2)-invariant 2-forms. Prove that V is finite-dimensional.

Proof by Arpan. Consider the Hodge star operator with respect to the Riemannian metric on M.

(1) A closed anti-self-dual form is harmonic.
(2) The space of harmonic forms is finite-dimensional.
(3) An SU(2)-invariant closed form is anti self-dual.
  (a) $\Lambda^+ = \operatorname{span}(\omega_I, \omega_J, \omega_K)$. M is a manifold of real dimension 4, which means that $\dim \Lambda^2(M) = 6$ and thus $\dim \Lambda^+(M) = 3$ (here we can check using a basis of $\mathbb{R}^6$ that $*$ permutes the basic covectors with no fixed points, so that it has zero trace; in turn this means that $\dim \Lambda^+ + \dim \Lambda^- = 0$). Further, each of ω_I, ω_J and ω_K squared and multiplied by $\frac{1}{2}$ give the Riemannian volume form, and therefore the three are in $\Lambda^+(M)$. Clearly they are linearly independent and thus span $\Lambda^+(M)$.
  (b) ω_I, ω_J and ω_K are not fixed by the SU(2) action Indeed, $J\omega_I = -\omega_I$ and similarly for other cases. Therefore there are no SU(2)-fixed points in Λ^+.
  (c) $\mathrm{SU}(2)\Lambda^+(M) = \Lambda^+(M)$. This is clear since $J\omega_I = -\omega_I$ and so on. Since $*$ is an isomorphism of $\Lambda^+(M)$ it follows that $\mathrm{SU}(2)\Lambda^-(M) = \Lambda^-(M)$.

  We conclude that if α is SU(2)-invariant (i.e. a fixed point of $\mathrm{SU}(2)\curvearrowright\Lambda^2$) its positive part will vanish, so that α is anti-self-dual. More precisely, if $\alpha = \alpha^+ + \alpha^-$ is SU(2)-invariant and $A \in \mathrm{SU}(2)$,

$$\alpha^+ + \alpha^- = \alpha = A\alpha = A\alpha^+ + A\alpha^-,$$

  so $A\alpha^+ = \alpha^+$, but SU(2) has no fixed points in Λ^+, so $\alpha^+ = 0$. Therefore $\alpha \in \Lambda^-$ as required.

□

Exercise 14.2. Let (M, K, J, K, g) be an almost hypercomplex Hermitian structure on a K3 surface, and $\omega_I, \omega_J, \omega_K$ its fundamental forms.

Suppse that ω_I is closed. Prove that ω_J, ω_K are closed, or find a counterexample.

The idea of the following solution (provided entirely by ChatGPT) is that we can "rotate" (in the J, K-plane within $\mathrm{End}(TM)$) two of the complex structures while preserving the quaternionic relations. Any orthonormal triple of such endomorphisms should be expected to satisfy the quaternionic relations (algebra), but the closedness condition on the associated fundamental forms will be modified (differential).

Proof. Let (M, I, J, K, g) be a hyperkähler structure on a K3 surface. Define

$$\begin{aligned}\tilde{J} &= \cos\theta J + \sin\theta K\\ \tilde{K} &= -\sin\theta J + \cos\theta K\end{aligned}$$

Then for any $v \in TM$,

$$\begin{aligned}\tilde{J}^2 v &= \cos J(\cos\theta Jv + \sin\theta Kv) + \sin\theta K(\cos Jv + \sin Kv)\\ &= \cos^2 J^2 v + \cos\theta\sin\theta JKv + \sin\theta\cos\theta KJv + \sin^2\theta K^2 v\\ &= -\mathrm{Id}v\end{aligned}$$

And likewise $\tilde{K}^2 = -\mathrm{Id}$. Also,

$$\begin{aligned}I\tilde{J}\tilde{K}v &= I(\cos\theta J(-\sin\theta Jv + \cos\theta Kv) + \sin\theta K(-\sin Jv + \cos\theta Kv))\\ &= I(-\cos\theta\sin\theta J^2 v + \cos^2\theta JKv - \sin^2\theta KJv + \sin\theta\cos\theta K^2 v)\\ &= -\cos\theta\sin\theta IJ^2 v + \cos^2\theta IJK - \sin^2\theta IKJv + \sin\theta\cos\theta IK^2 v\\ &= -\cos^2 v - \sin^2 v = -\mathrm{Id}v.\end{aligned}$$

Thus $(I, \tilde{J}, \tilde{K})$ is an almost hypercomplex structure on M. However,

$$\begin{aligned}d\omega_{\tilde{J}} &= d(g(\cdot, \tilde{J}\cdot))\\ &= d(g(\cdot, \cos\theta J\cdot + \sin\theta K\cdot))\\ &= d(\cos\theta\omega_J + \sin\theta\omega_K)\\ &= d\cos\theta \wedge \omega_J + d\sin\theta \wedge \omega_K\\ &= -\sin\theta d\theta \wedge \omega_J + \cos\theta d\theta \wedge \omega_K\\ &= d\theta \wedge \omega_{\tilde{K}},\end{aligned}$$

which is not always zero if θ is not constant. □

15. Instantons

Let X be a compact 4-dimensional compact oriented Riemannian manifold, G a compact Lie group (e.g. $U(n), \mathrm{SO}(n), Sp(n)$).

Definition 15.1. A G-*instanton* on X is a G-vector bundle E with a connection ∇ such that its curvature F_∇ is self-dual, i.e. $*F_\nabla = F_\nabla$ where $*$ is Hodge operator.

ADHM studied the case when $X = S^4$, and $X = \mathbb{C}P^2$ was also studied in a paper by N. Buchdahl.

Now we review the twistor space construction. Note that since X is 4-dimensional, the Hodge star operator is of the form $* : \Lambda^2 \to \Lambda^2$

16. Twistor space

Let X be a compact oriented 4-dimensional Riemannian manifold. Recall that since the Hodge star operator $*: \Lambda^2 \to \Lambda^2$ has eigenvalues ± 1, we have a decomposition

$$\Lambda^2 = \Lambda^+ \oplus \Lambda^- \cong \mathrm{SO}(T_X)$$

Consider the spherization of Λ^+:

$$Z := S\Lambda^+ = \{S \in \Lambda^+ : \text{unit vectors}\}$$

$(m, s) \in Z$, $m \in X$, $s_m \in \Lambda^+_m$,

$$T_{(m,s)}Z = T_m X \oplus_{LC} T_s S\Lambda^+_m \cong T\mathbb{C}P^1$$

which has a complex structure $\mathcal{I}_{\mathbb{C}P^1}$ (metric orientation).

Theorem 16.1. *$(Z, \mathcal{I}_Z = (s, \mathcal{I}_{\mathbb{C}P^1}))$ is a complex manifold of complex dimension 3.*

We also have a projection $p : Z \to X$ whose fibers are $\mathbb{C}P^1$, called *real lines*.

Example 16.2.
- $X = S^4$ then $Z = \mathbb{C}P^3$ (projective).
- $X = \mathbb{C}P^2$ then $Z = \mathbb{F}_3(\mathbb{C})$ (projective).

Let Z be a 3-dimension complex manifold. Let $L \hookrightarrow Z$ be a rational curve with a normal bundle $N_L \cong \mathcal{O}(1) \oplus \mathcal{O}(1)$.

Remark 16.3. If $Z \supset W$ where W is a complex submanifold, the small deformations of W are encoded by $H^0(N_W)$ and $H^1(N_W)$.

Let $\mathcal{M}$ be the sapce of deformations of L. Since $H^1(N_L) = 0$ then $TM \simeq H^0(N_L) = H^0(\mathcal{O}(1) \oplus \mathcal{O}(1))$, so $\dim_{\mathbb{C}} TM = 4$.

Let $\sigma : Z \to Z$ be an antiholomorphic involution with no fixed points but L (L is fixed). There is an induced antiholomorphic involution $\tilde{\sigma}$ on $\mathcal{M}$.

Remark 16.4.
- We can recover X as a connected component of the fixed point set of $\tilde{\sigma}$ in $\mathcal{M}$.
- We can recover the conformal strucutre.

Now we discuss Ward-Penrose transform. Let E be a G-instanton (i.e. a vector bundle with a connection such that $*F_\nabla = F_\nabla$) on X. Let $\phi : V \to \overline{V}^*$ be an Hermitian form.

There is a correspondence (the *Ward correspondence*) which allows to construct a holomorphic vector bundle on Z such that $\tilde{E}$ restricted to the fiber of $p : Z \to X$ (the fibers of p are $\mathbb{C}P^1$) trivially.

Conversely, starting from a hermitian holomorphic vector bundle $\tilde{E}$ on Z which restricts trivially to the fibers of $p : Z \to X$.

Problem (ADHM). Describe all G-instanton bundles on S^4 with $r \geq 2$, $c_2 = k$ (fixed second Chern class).

This makes us want to study the vector bundles on $\mathbb{C}P^3 = Z$.

See Horrseks for the original construction of monads. Upshot: "All bundles can be represented by the cohomology of some monad".

Definition 16.5 (Monad)**.** Let U, V, W be complex vector spaces over $\mathbb{C}P^1$. Consider the sequence $h \circ a = 0$, where $a, b$are bundle homomorphisms:

$$0 \longrightarrow U(-1) := U \otimes \mathcal{O}(1) \xrightarrow{a} V \otimes \mathcal{O} \xrightarrow{b} W(1) := W \otimes \mathcal{O}(1) \longrightarrow 0$$

Let $E := \frac{\operatorname{Ker} b}{\operatorname{Im} a}$.

Let's study the case of $\mathbb{C}P^3$. If Z_i are homogeneous coordinates on $\mathbb{C}P^3$,

$$a = \sum_i A_i Z_i, \qquad b = \sum_j B_j Z_j$$

the *monad condition* is

$$B_j A_i + A_j B_j = 0$$

[which gives the conditions for a hyperkähler metric on the moduli space].

Theorem 16.6 (Barth, Barth-Hulek)**.** *Let E be a holomorphic vector bundle of rank $r \geq 2$ on $\mathbb{C}P^3$ with a $c_2 = k$ and $E|_L$ =trivially. Then E is the cohomology of a monad. Also, it satisfies some vanishing conditions:*

$$H^0(E) = H^0(E^*) = H^1(E(-2)) = H^1(E^*(-2)) = 0.$$

17. Stability

This section outlines some concepts from a paper by Verbitsky from approx. 2002.

A *Gauduchon metric* is such that $\partial\bar{\partial}\omega^{n-1} = 0$. The degree of a holomorphic vector bundle is given by

$$\deg_g(\mathcal{F}) = \int_M c_1(\det \mathcal{F}) \wedge \omega^{n-1},$$

which is well defined when g is Gauduchon.

Definition 17.1. The *slope* of a vector bundle is

$$\mu_g(\mathcal{F}) = \frac{\deg_g \mathcal{F}}{\operatorname{rk}\mathcal{F}}.$$

A vector bundle $\mathcal{F}$ is *stable* if for any coherent subsheaf $\mathcal{E} \subset \mathbb{F}$

$$\mu_g(\mathcal{E}) < \mu_g(\mathcal{F}).$$

It is called *semistable* if we put $\leq$ instead.

Example 17.2. Line bundles are stable.

Theorem 17.3 (Hitchin-Kobayashi correspondence)**.** *Let (M, g) be a compact complex manifold with g Gauduchon. Then an hermitian vector bundle (F, h) is Hermitian-Einstein, i.e. it satisfies*

$$\sqrt{-1}\Lambda\Theta = \gamma id$$

if and only if it is polystable, i.e. it is the sum of stable bundles with the same slope.

The aim of Misha's paper is to classify stable bundles in positive elliptic fibrations using Hitchin-Kobayashi correspondence. Positive elliptic fibrations are $\pi : M \to X$ with an action of a torus on the fibers, and such that $\pi^*\omega_X = d\eta$ where ω_X is a Kähler form on X. Such a metric is called preferred if it is invariant on the fibres and is a Riemannian submersion.

The main theorem is the following:

Theorem 17.4. *For a positive elliptic fibration $\pi : M \to X$, if (B, h) is a stable bundle on M,*

$$B \simeq L \otimes \pi^* B_0$$

where L is a line bundle and B_0 is stable bundle on X.

The upshot is that a locally conformal hyperkähler manifold has an associated elliptic fibration…

18. Deformation theory (introduction)

Here I discuss deformations in the category of complex spaces.

Let's just do a summary of [Voi02, Chapter 9].

Let $\mathcal{X}$ be a complex manifold, B a complex manifold and $\phi : \mathcal{X} \to B$ a holomorphic map. We call ϕ a *family* if it is a proper holomorphic submersion.

A theorem by Ehresmann says that any "deformation" of smooth real manifolds is trivial:

Theorem 18.1 (Ehresmann)**.** *If $\phi : \mathcal{X} \to B$ is a proper submersion of differentiable manifolds (no complex structure), with B contractible and $0 \in B$ some distinguished point, then the total space $\mathcal{X}$ is trivial as a manifold over B, i.e. there exists a diffeomorphism*

$$T : \mathcal{X} \xrightarrow{\cong} X_0 \times B$$

of manifolds over B, i.e. commuting with projection.

Proof. The main tool is the theorem of tubular neighbourhoods, which says that there is a diffeomorphism of a nieghbourhood W of X_0 in $\mathcal{X}$ and a neighbourhood of X_0 in its normal bundle. (We have a normal bundle because X_0 is a submanifold of $\mathcal{X}$.)

In particular this gives a differentiable retraction $T_0 : W \to X_0$, which allows us to consider the map

$$(T_0, \phi) : W \to X_0 \times B.$$

Since T_0 is a retraction (it's the identity when restricted to X_0) and ϕ is a submersion, the map (T_0, ϕ) has nonsingular differential along X_0. Then we can invert it locally along X_0, and since X_0 is compact we obtain a whole neighbourhood $W' \subset W \subset \mathcal{X}$ where the differential is nonzero and thus $(T_0, \phi)|_{W'}$ is an embedding (using the inverse function theorem).

Finally we restrict the image of $(T_0, \phi)|_{W'}$ to a neighbourhood $U \subset B$ of 0, using properness of ϕ to ensure that $\phi^{-1}(U) \subset W'$ — though I'm not exactly sure how. We have constructed a diffeomorphism $\phi^{-1}(U) \to X_0 \times U$ commuting with the projection.

This proof was only local; the global version is in [Voi02] though it's not used in the rest of the text. □

In the complex situation, if $\phi : \mathcal{X} \to B$ is a family of complex manifolds and $0 \in B$, then "up to replacing B by a neighbourhood of 0", there exists a smooth trivialization $T = (T_0, \phi) : \mathcal{X} \to X_0 \times B$ such that the fibres of T_0 are complex submanifolds of $\mathcal{X}$.

Now let $\phi : \mathcal{X} \to B$ be a family of complex manifolds. Recall the pullback bundle $\phi^* B$ and the fact that the differential ϕ_* can be thought of as a gadget that maps tangent vectors to $\mathcal{X}$ not to vectors tangent to B but to vectors in the pullback bundle ϕ^*B. Restricting everything to the fibre $X = \phi^{-1}(0)$ we get

$$0 \longrightarrow T_X \longrightarrow T_{\mathcal{X}|X} \longrightarrow \phi^* T_{B|X} \longrightarrow 0. \tag{18.1.1}$$

There's what seems to be a really nice result saying that $H^1(\mathcal{F})$ parametrizes isomorphisms classes of extensions of $\mathcal{F}$ by the trivial bundle, but for now let me just define it roughly as follows. The category of $\mathcal{O}_X$ is most surely abelian with enough injectives. This means that objects have associated complexes, and we can take cohomologies of these complexes. This says, roughly, that short exact sequences of vector bundles have associated cohomology long exact sequences. Then the Kodaira-Spencer map is the map

$$\rho : T_{B,0} = H^0(X, \phi^* T_{B|X}) \to H^1(X, T_X)$$

induced by the long exact sequence associated to 18.1.1. Here the equality $T_{B,0} = H^1(X, \phi^* T_{B|X})$ is just because the pullback of the tangent bundle of B to any fiber is trivial.

Now I will explain first order deformations. Consider a family $\mathcal{X} \to B$. We introduce a complex analytic subscheme B_ε of B giving an ideal sheaf $\mathcal{M}_0^2$ where $\mathcal{M}_0$ is the maximal ideal of $\mathcal{O}_{B,0}$ consisting of the holomorphic functions vanishing at 0. This means that a function in $\mathcal{M}_0^2$ is (a finite sum of terms) of the form $f = gh$ for $g, h \in \mathcal{M}_0$, which means that $df = d(gh) = hdg + gdh = 0$ at 0 since both g and h vanish at 0. That is, this sheaf contains functions that vanish at 0 and whose first derivative vanishes at 0.

This means that B_ε is a point: it's the vanishing locus of the set of functions that vanish at a single point. However, it is endowed with an unusual structure sheaf. Let v be a tangent vector to B at 0 and f a function in the strange sheaf $\mathcal{M}_0/\mathcal{M}_0^2$. Then by the computation above, since v is a differential operator, v annihilates f. This means that this sheaf detects not only the point 0 but also any tangent vector at 0. Let's denote $B_\varepsilon = \operatorname{Spec} \mathbb{C}[\varepsilon]/(\varepsilon^2)$ just because its structure sheaf is $\mathbb{C}[\varepsilon]/(\varepsilon^2)$.

In fact, this is so throughout deformation theory: $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ is topologically a point! But it's a *fat point*: the support of its structure sheaf is a point, but as a scheme, it's not a point. And the same happens with the family $\mathcal{X}$ over $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$, topologically it's just the fibre X! But its structure sheaf is different: "it contains the directions in $\mathcal{X}$ which are normal to X".

Perhaps the most important part of this construction comes when considering a fat point $B_\varepsilon \subset B$. This gives a first-order deformation $\phi^{-1}(B_\varepsilon)) \subset \mathcal{X}$

$$\begin{tikzcd} X \arrow[r] \arrow[d] & \phi^{-1}(B_\varepsilon) \subset \mathcal{X} \arrow[d] \\ * \arrow[r] & B_\varepsilon \subset B \end{tikzcd}$$

Using the constructions above, one shows that to any such deformation we can associate a cocycle in $H^1(X, T_X)$. Thus, first order deformations are "exactly parametrised" by the group $H^1(X, T_X)$. This is the main idea behind Kodaira-Spencer map, which is well-explained in Wikipedia.

I will present a similar construction to Kodaira-Spencer map that works for embedded deformations instead of just deformations. (That's with the intention of understanding the forthcoming exercise Exercise 19.3.)

Since we are in complex space category I follow [Voi02, p. 224] for the main setup, but this is also a mixture of [**?**, p. 147, 150].

The result is in fact quite reasonable geometrically: the dimension of directions in which we can deform a submanifold should be non other than the dimension of the normal bundle.

19. Deformation theory

Definition 19.1. Let X be a complex manifold. A *deformation* of X is a holomorphic submersion $\pi : \mathcal{X} \to B$ of complex manifolds such that $\pi^{-1}(0) \cong X$ for some $0 \in B$.

If $\mathcal{X}$ and $\mathcal{Y}$ are two deformations of X over the same base, a *morphism of deformations* is a map $f : \mathcal{X} \to \mathcal{Y}$ such that the diagram

$$\begin{tikzcd} & X \arrow[dl] \arrow[dr] & \\ \mathcal{X} \arrow[rr, "f"] \arrow[dr] & & \mathcal{Y} \arrow[dl] \\ & B & \end{tikzcd}$$

commutes. The *space of deformations* of X is the set of deformations modulo the equivalence relation of two deformations being isomorphic. A priori we don't know if this set has any structure.

Let Y be another complex manifold. A *deformation of X in Y* is a deformation of X such that $\mathcal{X} \subset Y \times B$.

The *fat point* is a ringed space $(B_\varepsilon, \mathbb{C}[\varepsilon]/(\varepsilon^2))$ whose underlying topological space B_ε is a point and structure sheaf is the ring of dual numbers $\mathbb{C}[\varepsilon]/(\varepsilon^2)$.

A *first order deformation* of X is a deformation such that the base B is a fat point.

The following proposition is a mixture of ideas from [Voi02, p. 224] and [Ser06, Examples 3.2.4(ii)].

Proposition 19.2. *Let X be a smooth hypersurface in a complex manifold Y. For every deformation of X in Y with base of dimension 1 we can construct a section of the bundle $N_{X/Y}$. Different deformations give different sections up to rescaling.*

Proof. Let

$$\begin{tikzcd} X \arrow[r] \arrow[d] & \mathcal{X} \subset Y \times B \arrow[d, "\pi"] \\ 0 \arrow[r] & B \end{tikzcd}$$

be a deformation of X in Y and suppose that $\dim B = 1$. This means that $\mathcal{X}$ is a complex submanifold of the product $Y \times B$ and $\pi : \mathcal{X} \to B_\varepsilon$ is a holomorphic submersion.

Since X is a hypersurface, we can find an open cover V_i of M such that $V_i \cap X := U_i$ is given locally by the vanishing of a function $f_i \in \mathcal{O}_Y(V_i)$. In terms of coordinates we find that that $V_i \cap X$ is given by $(z_1, \ldots, z_{n-1}, 0)$.

Since π is a submersion, by the holomorphic inverse function theorem we obtain for every $V_i \subset \mathcal{X}$ (such that $V_i \cap X \neq \emptyset$) a biholomorphism from some $B_i \subset B$

containing 0 to V_i. This gives coordinates $(0,\dots,b)$ to V_i. We conclude that $V_i \cong U_i \times B_i$ along X.

Now we "restrict" to a first order deformation. Consider the fat point subscheme B_ε of B (I think this is the reason why B must have dimension 1, in Voisin notation, that a first order neighbourhood of 0 in B is indeed a fat point) and let $X_\varepsilon = \pi^{-1}(B_\varepsilon)$, which is a ringed space whose topological space is the same as X, but its structure sheaf is induced from B_ε.

(I think that to prove this part I have to look carefully at the induced sheaf $\pi^*(\mathbb{C}[\varepsilon]/(\varepsilon^2)$.) Since we are on a local trivialization, the structure sheaf of the product $U_i \times B_i$ is the tensor product of the structure sheaves (reference?), so that the structure sheaf of X_ε at $U_i \times B_\varepsilon$ is given by

$$\mathcal{O}_{X_\varepsilon}(U_i) = \mathcal{O}_Y(U_i) \otimes \mathbb{C}[\varepsilon]/(\varepsilon^2).$$

Functions here look like finite sums of functions of the form $f + \varepsilon g$ for $f, g \in \Gamma(U_i, \mathcal{O}_{U_i})$.

The next key step is to argue that X_ε is locally cut by functions of the kind $f_i + \varepsilon g_i$ for some $g_i \in \mathcal{O}_M(U_i)$. Here the f_i are the same as those which define X. I think a proof is that from the ideal sheaf exact sequence of X in Y

$$0 \longrightarrow \mathcal{I}_X \longrightarrow \mathcal{O}_Y \longrightarrow \mathcal{O}_X \longrightarrow 0$$

we obtain after tensoring by $\mathbb{C}[\varepsilon]/(\varepsilon^2)$

$$0 \longrightarrow \mathcal{I}_X \otimes \mathbb{C}[\varepsilon]/(\varepsilon^2) \longrightarrow \mathcal{O}_Y \otimes \mathbb{C}[\varepsilon]/(\varepsilon^2) \longrightarrow \mathcal{O}_X \otimes \mathbb{C}[\varepsilon]/(\varepsilon^2) \longrightarrow 0.$$

(But is the functor $-\otimes \mathbb{C}[\varepsilon]/(\varepsilon^2)$ exact? Why? So far we didn't impose a flatness condition on π.) The sheaf in the right coincides with the structure sheaf of X_ε when evaluated on a trivial open set of the form $U_i \otimes B_\varepsilon$. This shows that the ideal sheaf of X_ε as a ringed space is $\mathcal{I}_X \otimes \mathbb{C}[\varepsilon]/(\varepsilon^3)$. Since $\mathcal{I}_X$ is locally principal generated by f_i, we conclude that indeed X_ε must be cut locally by functions of the form $f_i + \varepsilon g_i$ for some $g_i \in \mathcal{O}_Y(U_i)$

Recall that the line bundle associated to X is $\mathcal{O}_Y(X)$, given by the cocycle $f_i/f_j := f_{ij}$. A version of adjunction formula (more precisely, [GH78, Adjunction formula I, p. 146]) establishes that this line bundle, when restricted to C, is in fact the normal bundle of X in Y, defined as que quotient bundle $T'_Y/T'_X := N_{X/Y}$. In symbols, $\mathcal{O}_Y(X)|_X \cong N_{X/Y}$.

Similarly, the line bundle associated to $\mathcal{X}$ is given by the cocycle

$$F_{ij} = f_{ij} + \varepsilon g_{ij}$$

for $g_{ij} = \frac{g_i - f_{ij} g_j}{f_j}$. Indeed, note that $\frac{1}{f_j + \varepsilon g_j} = \frac{1}{f_j} - \varepsilon \frac{g_j}{f_j^2}$ and compute $F_{ij} = (f_i + \varepsilon g_j)/(f_j + \varepsilon g_j)$.

Then

$$f_i + \varepsilon g_i = (f_{ij} + \varepsilon g_{ij})(f_j + \varepsilon g_j)$$

gives

$$g_i = f_{ij} g_j + g_{ij} f_j$$

and since f_j vanishes along X, we are left with $g_i = f_{ij} g_j$ which is how a section of the sheaf $N_{X/Y}$ is obtained. □

Exercise 19.3. Let C be a smooth genus g curve which can be embedded in a K3 surface M, and $\mathcal{X}$ "the family of all deformations" of C in M. Prove that $\dim \mathcal{X} \le g$.

Proof. By Adjunction formula II (Theorem 7.6) and M being a K3 surface we get that $h^0(C, N_{C,M}) = g$.

If the deformation space X exists and is a complex manifold, the map constructed in Proposition 19.2 would be a holomorphic (why holomorphic!?) injective map $X \to H^0(C, N_{C/Y})$. This obliges $\dim X \leq h^0(C, N_{C/Y}) = g$. Indeed, if not, the differential must have a kernel, and integrating along a direction in the kernel we obtain the germ of a complex curve along which the map is constant, which is impossible since it is injective. □

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